An Empirically Informed Theory of Paradigmatic Change: Evidence from Media Coverage of Climate Change in Canada

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Abstract

This study develops an empirically-informed theory of paradigmatic change through analysis of climate discourse in Canadian news media. Drawing on the Canadian Climate Framing (CCF) database, comprising over 260,000 articles from 20 newspapers (1978-2025) annotated across 65 categories, we introduce novel methods to identify and track paradigmatic shifts in media framing. While existing research identifies dominant frames abstractly, we operationalize paradigms as specific configurations of frames that persistently structure discourse. Our multi-method approach combines information theory, network analysis, causality testing, and proportional dominance metrics to determine which frames achieve hegemonic status. Results reveal that Canadian climate discourse transitioned from a mono-frame paradigm in the early 1980s to a stable multi-frame paradigmatic structure. The political frame emerges as unequivocally dominant, followed by the economic frame, while the scientific frame, though dominant in early years, shows declining centrality over time. Year-by-year analysis (1978-2024) demonstrates that triple-frame configurations are most common, with the political-economic dyad increasingly conditioning the presence of other frames. This shift from science-centered to politically and economically dominated discourse reflects the mainstreaming and "economization" of climate change. Our framework provides the first systematic empirical approach to measuring paradigmatic change in media discourse, with implications for understanding how dominant interpretive structures shape policy debates.

1 Main

"The deliberation of public policy takes place within a realm of discourse. Policies are made within some system of ideas and standards" (Hall 1993, p. 279). These systems of ideas are *paradigms*, overarching frameworks that specify not only the goals of policy and the instruments used to achieve them, but also the very nature of the problems they are meant to address (Hall 1993). According to Skogstad (2011), paradigms function as shared interpretive frameworks that become institutionalized through repeated use, shaping how actors understand reality and define appropriate responses. While Hall's seminal work examined economic policy paradigms, the concept has broader applicability to understanding how societies interpret and respond to complex issues. Yet fundamental questions remain underexplored: What concretely constitutes a paradigm? How can it be empirically measured, observed, and studied? Under what conditions does it change?

Drawing on climate change as a case study and a comprehensive database of more than 260,000 media articles (see Appendix A), we argue that paradigms are composed of multiple frames. Frames are distinct interpretative lenses through which phenomena are understood. In media discourse, frames represent different ways to define problems, attribute causality, and imply solutions by "selecting some aspects of a perceived reality and making them more salient" (Entman 1993, p. 52). However, we argue that not all frames contribute equally to a paradigm; only the dominant ones. For instance, climate change can be framed as a scientific problem, a political challenge, an economic concern, a matter of social justice and so on; but only the frames that consistently dominate discourse over time will define the hegemonic paradigm. This article develops an empirical approach to identify which frames achieve dominance, how they combine to form paradigms, and under what conditions these paradigmatic configurations shift.

The way media covers climate change matters. It influences both public opinion and the political agenda through two key mechanisms: agenda-setting, by making issues more salient (McCombs and Shaw 1972), and framing, by shaping how issues are interpreted and what responses are seen as appropriate (Chong and Druckman 2007). Scholars studying climate coverage have documented temporal fluctuations in attention (Schmidt, Ivanova, and Schäfer 2013; Stoddart and Smith 2016; Hase et al. 2021), following non-linear trajectories consistent with Downs (1972) issue attention cycle. They have also identified multiple frames through which climate change is portrayed, with certain frames appearing more frequently than others (Nisbet 2009; Badullovich, Grant, and Colvin 2020). However, what precisely constitutes a "dominant" frame remains empirically and theoretically underdeveloped. Studies typically designate frames as dominant based solely on proportional prevalence, without systematic empirical criteria or consideration of how frames interact to form broader discursive structures, that is, *paradigms* (O'Neill et al. 2015; Stoddart and Smith 2016).

Building on this literature, we propose that paradigms emerge when certain frames become dominant and persistently structure discourse over time. In media discourse, this institutionalization occurs when journalists consistently mobilize particular combinations of frames, creating stable patterns of problem definition and solution framing. A *hegemonic paradigm* thus consists of the specific configuration of dominant frames that prevails in a given period. Not a single interpretive lens, but a constellation of mutually reinforcing perspectives that collectively shape understanding. When certain frames consistently co-occur and reinforce each other, they form the paradigmatic structure that governs how the issue is understood and debated.

The Canadian Climate Framing (CCF) Dataset (Lemor, Pillod, and Taylor 2025), a comprehensive media corpus developed by the authors, represents the largest bilingual climate discourse database in Canada. Spanning 47 years (1978-2025), it comprises 266,271 systematically collected climate-related articles from 20 major Canadian newspapers and state-of-the-art machine learning annotations (see Appendix A). Over 9.2 million sentences are coded using 65 hierarchical categories organized into four primary dimensions—thematic frames (economic, health, security, justice, political, scientific, environmental, cultural), actors and messengers, events, and solutions; plus additional contextual variables including emotional tone, geographic focus, and urgency indicators. Each annotation was produced by transformer-based models (BERT for English, CamemBERT for French) trained on over 4,000 expert-annotated sentences, achieving a macro F1 score of 0.85.

Using observational data from the CCF dataset, this study develops empirical methods to measure and track the composition of climate change paradigms in Canadian media discourse. The overarching goal is to produce an empirically-informed theory of paradigmatic shifts in climate change coverage. We first propose a method to measure whether the climate paradigm's internal structure is mono-frame (dominated by a single interpretive lens) or multi-frame (composed of multiple co-existing frames). We then develop a comprehensive analytical framework to identify which specific frames achieve dominance within the overall discourse through a multi-method approach that combines information theory, network analysis, causality testing, and proportional dominance metrics. Finally, we trace the evolution of the hegemonic paradigm by applying these methods year-by-year to reveal how the constellation of dominant frames shifts over time and thus how the hegemonic paradigm of climate change evolves.

The subsequent stages of this research, currently in development, will complete our empirically-grounded theory of paradigmatic change. The second stage will develop a media frame cascading model to detect periods when journalists, outlets, and other actors converge on similar frames within specific temporal and geographic contexts, signaling potential paradigm transformation while accounting for the media's own institutional logic (Cook 2005; Esser and Strömbäck 2014). The third stage will examine how focusing events, such as scientific reports, climate summits, or large-scale protests, coincide with these cascades. While agenda-setting research has traditionally examined how such events affect issue salience (Kingdon 1984; Birkland 1997; Baumgartner and Jones 2009), we will investigate their impact on frame salience, hypothesizing that different types of focusing events trigger distinct framing responses that can catalyze paradigm shifts. Together, these stages will provide the first comprehensive empirical framework for understanding when and how paradigms change in media discourse.

2 Methods

2.1 Data Source

This study analyzes the Canadian Climate Framing (CCF) database, which contains 266,271 climate-related articles from 20 Canadian newspapers published between 1978 and 2025. The complete methodology for database construction, including data collection, preprocessing, and annotation protocols, is detailed in Appendix A. All analyses were performed on sentence-level annotations across eight primary thematic frames: Economic, Health, Security, Justice, Political, Scientific, Environmental, and Cultural.

2.2 Determining Paradigm Composition: Mono-frame versus Multi-frame Structure

To determine whether climate coverage in Canada follows a mono-frame or multi-frame paradigm structure, we employed Shannon entropy as our primary diversity metric, calculated on weekly frame proportions across the entire dataset.

The critical threshold separating mono-frame from multi-frame paradigms was established at $H_{normalized} = 0.333$, corresponding to $\log_2(2)/\log_2(8)$. This threshold represents the normalized entropy when two frames are equally distributed, marking the transition point between mono-frame dominance and multi-frame diversity. Above this threshold, the discourse necessarily involves meaningful contributions from multiple frames rather than being dominated by a single interpretive lens.

To operationalize this analysis, we aggregated sentence-level frame annotations to weekly proportions, calculating the average detection probability for each frame across all sentences published in a given week. Shannon entropy was then computed for these weekly frame distributions, normalized by $\log_2(8)$ to yield values between 0 (complete dominance by one frame) and 1 (equal distribution across all frames). To identify distinct paradigmatic periods, we applied a 12-week rolling median filter to smooth high-frequency fluctuations while preserving major structural shifts. Statistical significance was assessed through bootstrap analysis (1,000 iterations) to establish 95% confidence intervals for the median entropy, complemented by a Wilcoxon signed-rank test to determine whether the observed entropy values significantly exceeded the multi-frame threshold.

2.3 Measuring Frame Dominance Across the Corpus

Identifying which frames achieve dominance within a paradigm requires distinguishing between frames that merely appear frequently and those that fundamentally structure discourse. Since paradigms represent institutionalized patterns of understanding, dominant frames must demonstrate multiple forms of centrality, we consider that: (1) they must carry substantial information content that shapes interpretation of other frames, (2) they must occupy structurally central positions in the network of frame relationships, forming conceptual hubs around which other frames organize, (3) they must exhibit causal influence over the presence and absence of other frames, and (4) they must maintain consistent proportional presence across time. To capture these theoretically distinct dimensions of dominance, we develop a comprehensive multi-method approach integrating four equally-weighted analytical components, each contributing 25% to the final dominance score (see Appendix C for technical details).

The four analytical components evaluate complementary dimensions of frame dominance: information theory analysis measures how much each frame contributes to the overall information structure through mutual information and entropy reduction; network analysis identifies structurally central frames through PageRank and eigenvector centrality metrics computed on co-occurrence networks; causality analysis determines which frames drive others through Granger causality tests and transfer entropy calculations; and proportional dominance assesses sustained presence through mean proportions, temporal consistency, and elevation above baseline levels. Rather than imposing an arbitrary threshold, dominant frames are identified through a consensus approach that applies four statistical methods to the composite scores (see Appendix C for detailed specifications). A frame achieves dominant status only when selected by at least three of these four methods.

2.4 Analyzing Temporal Evolution of Frame Dominance

To trace how the hegemonic paradigm evolves over time and identify periods of paradigmatic stability versus transformation, we applied our dominance analysis methodology independently to each year from 1978 to 2024. This year-by-year approach allows the composition of the hegemonic paradigm to vary naturally according to each period's specific discursive dynamics rather than imposing a fixed threshold across all time periods. For each year, the complete four-component analysis was conducted on that year's 52-53 weeks of data, with dominant frames identified through the same consensus approach requiring selection by at least three of four statistical methods. This independent annual analysis helps to determine the number of dominant frames for each year, and thus the hegemonic paradigmatic structure. Additionally, we tracked paradigm concentration through Shannon diversity indices to quantify whether the discourse becomes more concentrated or diverse over time, while network evolution analysis examined how frame relationships restructure across overlapping decades.

3 Results

3.1 Mono-frame paradigm or multi-frame paradigm?

Analysis of the CCF dataset reveals that Canadian media discourse on climate change is predominantly characterized by a multi-frame paradigm. The observed median entropy value (0.72) lies well above the mono-frame threshold of 0.333, with strong statistical significance ($p < 0.0001$), indicating that multiple frames are consistently mobilized rather than reliance on a single dominant frame. The effective number of frames hovers around 4-5 throughout most of the period, suggesting that climate discourse typically draws upon the equivalent of five to six equally-weighted interpretive frameworks.

As shown in Figure 1, the early period from 1978 to the early 1980s exhibited values below the threshold which indicates a mono-frame paradigm. The transition occurred gradually, with entropy values crossing the threshold around 1982 and firmly establishing themselves above it by 1985. This marks a clear rupture and shift toward a multi-frame paradigm that has remained remarkably stable over the subsequent four decades. Consequently, only a brief initial period of media coverage was characterized by mono-frame dominance, while the vast majority of climate discourse has exhibited a multi-frame structure.

3.2 Which frames dominate coverage?

Our overall analysis (1978-2024) identifies only two frames as dominant in Canadian climate coverage through the stringent 3/4 consensus approach when analyzing the entire period as a whole: the political frame (0.932) and the economic frame (0.622) (Figure 2b)[1]. These two frames demonstrate structural dominance across all dimensions.

Figures 2c and 2d provide further details on the scores attributed to each frame across indicators. The political frame consistently achieves the highest score, with exceptional performance across all four analytical dimensions, which makes it unequivocally dominant. Figures 2a and 2d illustrate the political frame's dominance across specific indicators. Its average proportion of use (Figure 2a) shows that, since the late 1980s, it has increasingly become the most prevalent suggestion: predominant(?) frame. It also ranks highest in network centrality across time (Figure 2d), where it emerges as a major hub, indicating that its presence often conditions the presence of other frames. The economic frame, while also dominant, shows more variable performance across indicators, achieving strong scores in network and proportional dimensions but moderate causality scores.

3.3 How does the paradigm's composition change over time?

Temporal analysis of frame dominance reveals significant shifts in the paradigm's composition over the 47-year study period (Figure 3a). Each year from 1978 to 2024 was analyzed independently using the identical four-method composite scoring system described above, with dominant frames determined through the 3/4 consensus approach described in the Methods. The scientific frame dominated the early period, particularly in the 1980s when

[1] The scientific frame (0.590) narrowly missed the dominance threshold, being selected by only two of the four statistical methods.

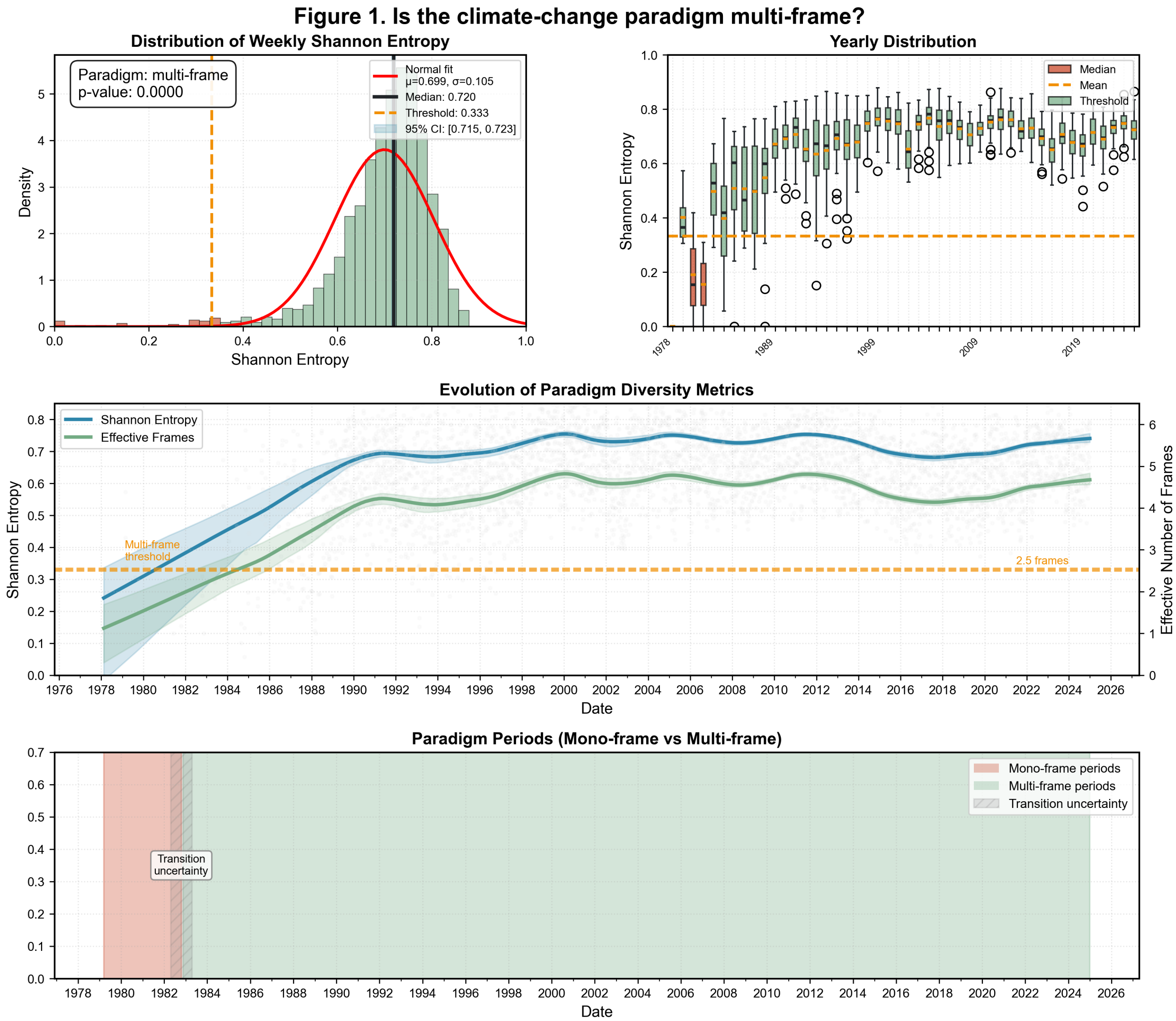

Figure 1: Is the climate-change paradigm multi-frame? Distribution of Weekly Shannon Entropy showing the evolution from mono-frame to multi-frame paradigm over time.

climate change first emerged as a public issue, and maintains a significant presence through the 1990s, though its relative importance gradually declines over time. The economic frame shows remarkable consistency across the entire period, while the political frame becomes increasingly dominant from the 1990s onward. This reveals a gradual shift from science-centered discourse in the early years to a persistent co-dominance of political and economic frames in recent decades, with science playing a declining but still significant role.

Figure 3b illustrates the evolution of the number of frames included in the paradigm over time. Triple-paradigm years (3 frames) represent the most common configuration across the 47 years analyzed. Dual-paradigm years (2 frames) are relatively rare overall, though they become more frequent after 2020. Quad-paradigm years (4 frames) occur sporadically throughout the period. The only mono-paradigm year was 1978, at the very beginning of climate coverage, dominated by the scientific frame. This distribution reveals that Cana-

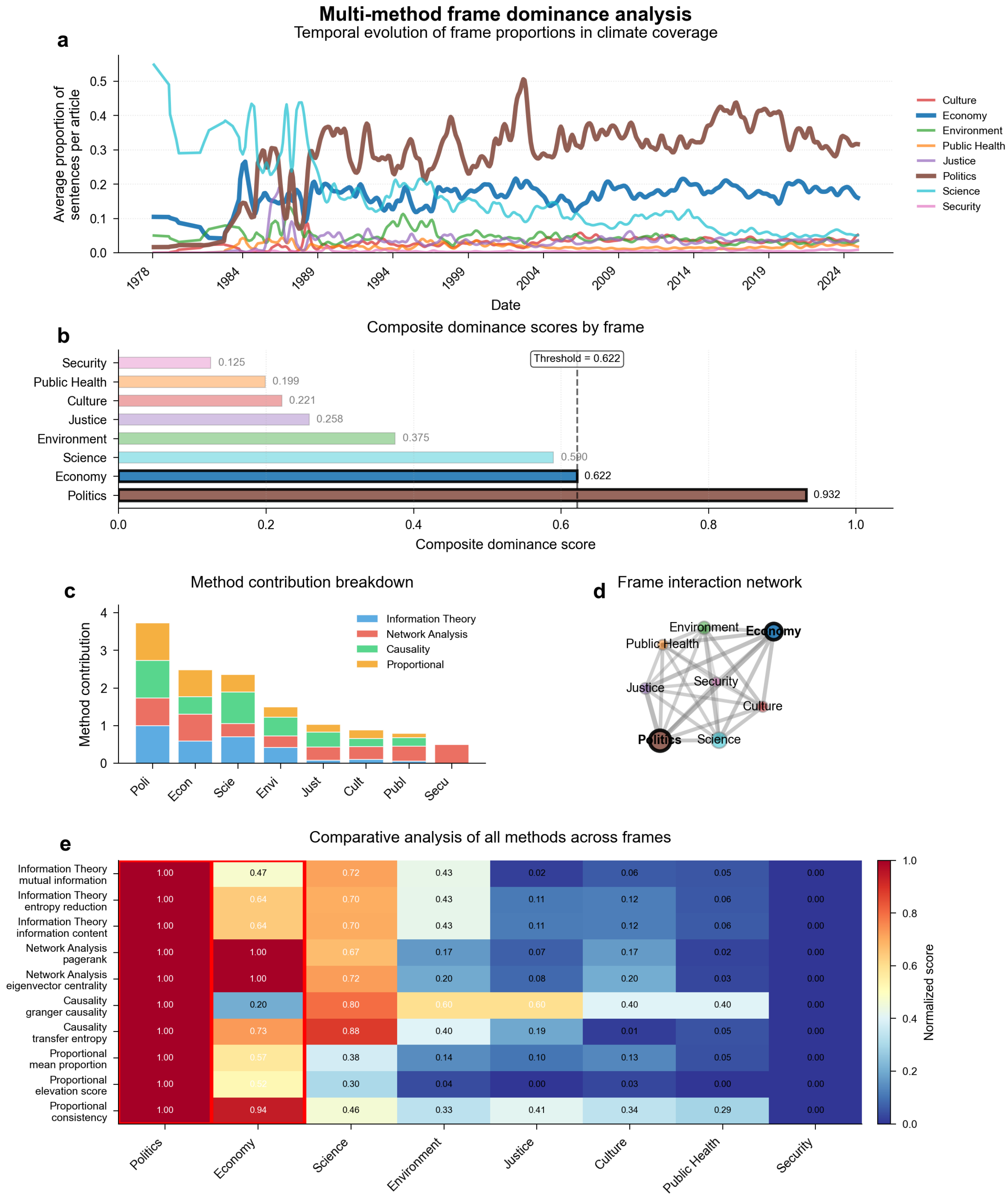

Figure 2: Multi-method frame dominance analysis. (a) Temporal evolution of frame proportions in climate coverage. (b) Composite dominance scores by frame. (c) Method contribution breakdown. (d) Frame interaction network. (e) Comparative analysis of all methods across frames.

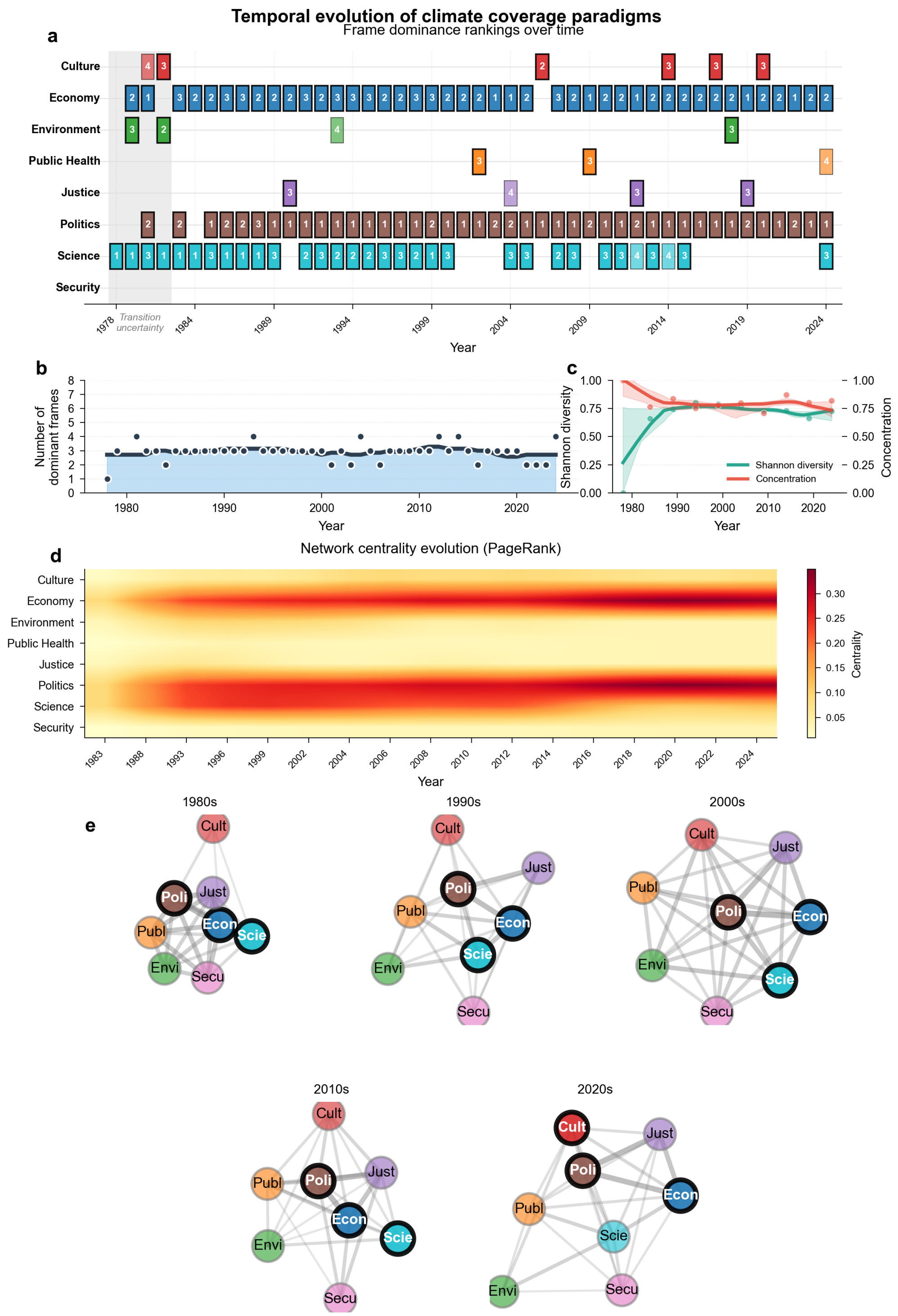

Figure 3: Temporal evolution of climate coverage paradigms. (a) Frame dominance rankings over time. (b) Number of dominant frames. (c) Paradigm diversity measured by Shannon diversity and concentration index. (d) Network centrality evolution measured by PageRank. (e) Frame co-occurrence networks across decades.

dian climate discourse has consistently mobilized multiple frames simultaneously, with three frames being the modal configuration across nearly five decades.

Figures 3c and 3d further highlight the degree of centrality of frames within the hegemonic paradigm. Shannon's entropy shows that the paradigm quickly diversified in the late 1980s and has remained relatively high. The concentration index shows fluctuations over time, with a notable increase in dual-paradigm years after 2020. The temporal dynamics reveal that while the political and economic frames have become increasingly central, the scientific frame continues to appear as dominant in many years through the 2010s. Other frames (Cultural, Public Health, Social Justice, Environmental) appear sporadically as dominant, which suggests that climate discourse occasionally expands to incorporate these perspectives, potentially during specific events or contexts.

4 Discussion

Although research on climate change coverage has expanded, few studies focus specifically on Canada. Moreover, existing work often relies on small samples, limited outlets, typically national and English-language, and short time spans. By contrast, the CCF includes over 260,000 articles from 20 newspapers, both French- and English-speaking, published since 1978, which enables a more fine-grained analysis of coverage.

Our results show that Canadian media coverage of climate change is largely characterized by a multi-frame paradigm, in which a limited but recurring set of frames is mobilized by journalists and newspapers. The fact that there is only one mono-frame period observed and that it is in the early years when the issue started to emerge on the public agenda (during the late 1970s and early 1980s) suggests that coverage has grown increasingly complex over time. Climate change can be thought of as a multidimensional, "wicked" problem, with far-reaching implications for domains such as the economy, defense, and justice, and with no straightforward solutions (Grundmann 2016). In that sense, the appearance of new frames appears to reflect a growing awareness of these broader societal impacts. This issue complexification has also been noted at a smaller scale by (Stoddart and Tindall 2015), who analyzed climate change coverage in the Globe and Mail and the National Post between 1999 and 2010.

When examining the frames that dominate media coverage with our stringent 3/4 consensus requirement, a nuanced picture emerges. In the overall analysis (1978-2024), only two frames, political and economic, meet the dominance threshold when analyzed across the entire period. However, the year-by-year analysis from 1978 to 2024 reveals greater complexity: the scientific frame dominates the earliest years and maintains dominant status in a majority of individual years, particularly through the 1980s and 1990s, though its presence declines over time. The economic frame shows remarkable consistency across the entire period, while the political frame becomes increasingly prevalent from the 1990s onward. This pattern suggests not an abrupt displacement of science, but rather a gradual layering where political and economic frames increasingly co-occur with and sometimes overshadow scientific framing. This subtle shift has been documented in other studies on Canada (Einsiedel 1992; Einsiedel and Coughlan 1993; Young and Dugas 2011) as well as internationally (e.g., (Pulver and and Sainz-Santamaría 2018)). The growing centrality of political and economic frames, confirmed by several studies (Young and Dugas 2011; Ahchong and Dodds 2012; Stoddart and Tindall 2015; Stoddart and Smith 2016), reflects the mainstreaming of climate change from a primarily scientific issue to one deeply embedded in political and economic discourse.

Furthermore, the decline of the scientific frame and the rise of the political frame appear to be closely connected. Research has shown that the coverage of climate change became increasingly politicized through the changing types of expertise mobilized by journalists, with scientists being progressively replaced by political actors as primary sources (Trumbo 1996; Chinn, Hart, and Soroka 2020). The emergence of the economic frame during the same period coincides with what many scholars describe as the "economization" of the issue, marked by the rise of sustainable development discourse, which presented economic growth and environmental protection as compatible goals. Hajer (2009) refers to this as "ecological modernization," a term he uses in a critical sense. The simultaneous rise of political and economic frames thus reflects the growing complexity of the issue. Notably, our use of innovative network centrality indicators reveals that these two frames have become increasingly central over time, to the point that the presence of other secondary frames is often linked to their coexistence. More broadly, network indicators provide new ways to examine how frames interact with one another, something that remains underexplored in the literature (Snow et al. 1986).

These preliminary findings are especially promising for the next stages of this study, notably the analysis of the impact of focusing events on frame salience. Many events of the 1980s and 1990s had long-lasting implications for how climate change is now defined and governed. More broadly, the hierarchy of frames in media coverage is significant because it can shape the range of policy solutions considered by decision-makers (McCombs and Shaw 1972; Gusfield 1981). If the public agenda shapes the political agenda, the decline of the scientific frame is especially concerning, as it suggests that climate policy development may increasingly be guided by factors other than scientific evidence.

5 Conclusion

In this study, we introduce new methodological tools to assess frame dominance by combining approaches from public policy studies and media studies. Echoing Wolfe, Jones, and Baumgartner (2013) call for closer collaboration between these fields in agenda-setting research, we argue that such integration is equally valuable for studying other media effects, particularly framing. While our application focuses on climate change coverage, the methods we propose are replicable and adaptable to a wide range of media topics. They offer scholars new ways to better capture frame dynamics, and future research could explore how shifts in frame dominance within media coverage feed back into and shape the policy process.

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A Complete Methodology: The Canadian Climate Framing (CCF) Database

A.1 Overview of the full methodology

The Canadian Climate Framing (CCF) project adopts a mixed-methods design that couples large-scale data collection with state-of-the-art machine learning (ML). It provides a systematic, reproducible framework for large-scale analysis of climate discourse that allows researchers to track how climate change is discussed, who participates, and how narratives evolve over time.

Figure 4 presents the complete methodological pipeline and illustrates how we transform raw newspaper articles into a richly annotated database suitable for climate communication research. The pipeline integrates human expertise with computational efficiency through an iterative human-in-the-loop machine learning procedure. This approach ensures both the scalability needed to process 266,271 climate-related articles and the accuracy required for rigorous academic research.

The methodology comprises four distinct phases: (1) the *Collection Phase* involves systematic gathering of climate-related articles from 20 major Canadian newspapers using carefully constructed Boolean search queries and resulting in 266,271 articles spanning 1978-2025; (2) the *Preprocessing and Design* phase transforms raw text into structured data, including sentence segmentation, language verification, and development of our comprehensive 65-category annotation framework; (3) the *Machine Learning Training* phase creates and validates transformer-based machine learning models (BERT for English, CamemBERT for French) through iterative training with over 4,000 expert manual annotations; and (4) the *Validation and Deployment* phase ensures rigorous quality assurance through performance metrics and manual verification, followed by application of trained models to annotate the complete corpus of 9.2 million sentences.

A.2 Phase 1 & 2: Data collection, preprocessing, database creation and annotation protocol

A.2.1 Step 1: Data collection

The Canadian Climate Framing (CCF) database was built through a systematic protocol to trace the evolution of climate discourse in Canada's bilingual media, reaching back to the earliest available archives. The corpus spans 47 years, from 1978 to the present, with 1978 marking the earliest article identified. To identify climate-related articles, we developed comprehensive Boolean search queries tailored to each language (Table 1). The queries included a range of terms capturing various dimensions of climate discourse to ensure inclusivity of different terminologies used. We selected 20 major Canadian newspapers to ensure broad geographic coverage across all provinces and a balanced representation of English- and French-language newspapers, the country's two official languages. These queries yielded an initial corpus of over 300,000 articles. Specifically, we searched the full text (title and main text) using the following terms:

Table 1. Boolean queries used for article retrieval

Language	Boolean Query
English	"global warming" OR "climate change" OR "climate disruption" OR "climate disturbance" OR "climate disturbances" OR "climate crisis" OR "greenhouse gas"
French	"réchauffement climatique" OR "réchauffement planétaire" OR "changement climatique" OR "changements climatiques" OR "dérèglement climatique" OR "crise climatique" OR "gaz à effet de serre"

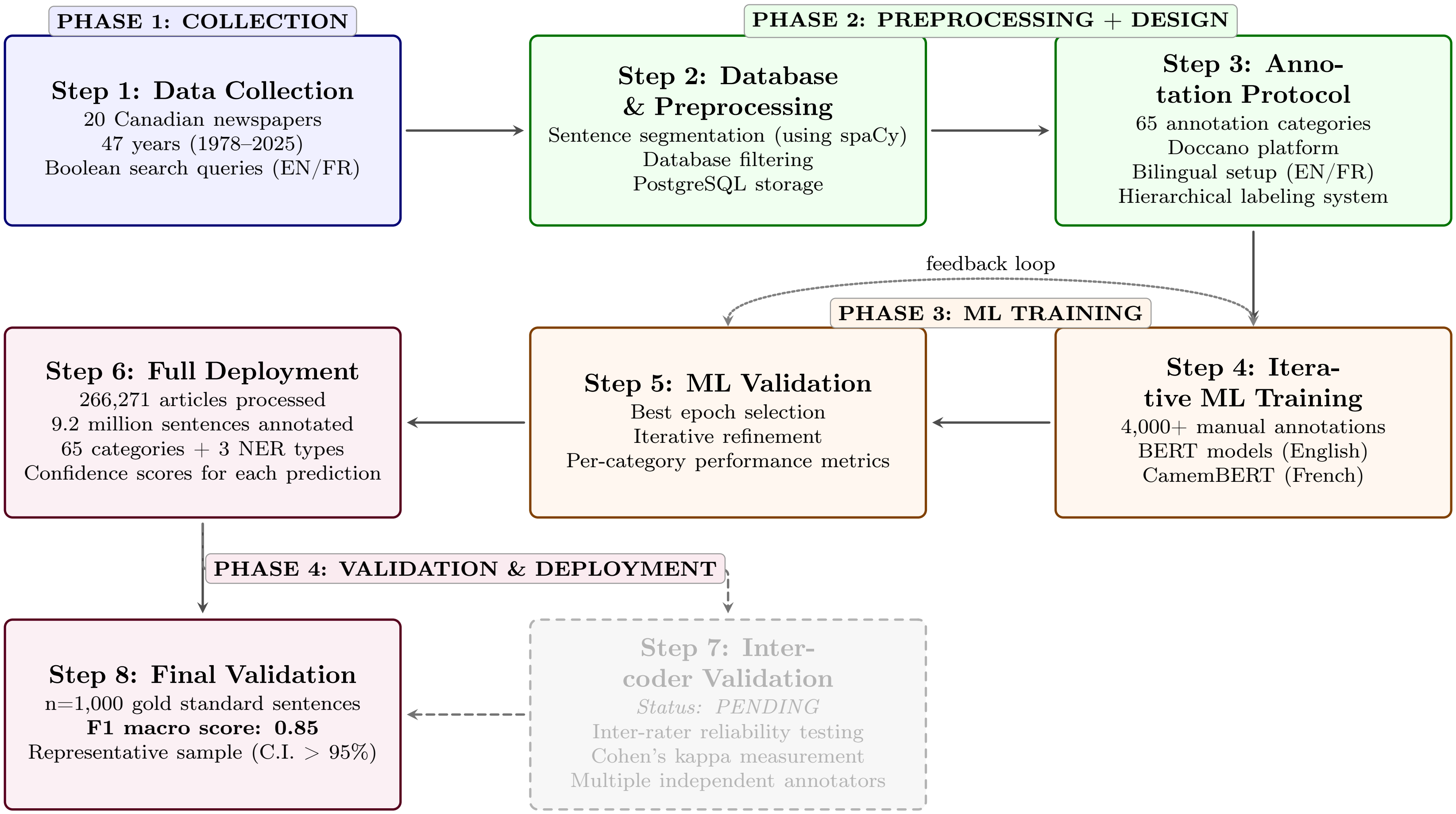

Figure 4: Methodological pipeline of the Canadian Climate Framing (CCF) project.

A.2.2 Step 2: Preprocessing

The preprocessing phase transformed this raw collection into a structured dataset suitable for machine learning analysis. Each article was segmented into two-sentence contexts using language-specific spaCy models[2]. Sliding windows with single-sentence overlaps were implemented to ensure complete coverage while maintaining semantic continuity throughout each article. The choice of two-sentence units was made to provide sufficient information in each segment, given the complexity of some categories we developed.

Quality control measures were applied systematically throughout the data preparation pipeline. We identified and removed duplicate articles with fuzzy string-similarity algorithms set to a 95% threshold. This process reduced the corpus to 266,271 unique articles[3]. We verified language with fastText[4] to ensure accurate categorization of bilingual content, and we standardized dates and performed UTF-8 normalization to resolve inconsistencies that stemmed from multiple data sources. This data preparation produced a clean, standardized corpus ready for annotation, which we stored in two PostgreSQL tables: one with full metadata (including titles, main text, authors, dates, and page numbers[5]) and another with the processed text data (including sentence segments and their corresponding IDs).

The resulting database is illustrated in Figure 5, which displays the distribution of article counts by media (20 outlets), while Figure 6 shows total article counts per year across the full time span (excluding 2025[6]). Finally, Figure 7 presents the distribution of articles by province, based on the primary location of each newspaper. This geographic breakdown shows the representativeness of the database across Canada (with 17.1% of French articles, and 82.9% of English articles). Three national newspapers are omitted from the provincial breakdown due to their pan-Canadian scope; nevertheless, they account for 36.2% of the articles: The Globe and Mail, the National Post, and the Toronto Star.

A.2.3 Step 3: Annotation protocol

The annotation framework represents the conceptual foundation of the CCF database. It encompasses 65 distinct categories organized into a hierarchical taxonomy. This comprehensive system was built through iterative refinement and combines established categories from climate communication literature with novel classifications identified through exploratory analysis of the CCF Database. The framework operates at three interconnected levels: detection-level binary classification determining the presence or absence of primary categories, sub-categorization providing fine-grained classification within detected categories, and

[2]We employed en_core_web_lg for English text and fr_dep_news_trf for French text, both optimized for news content processing.

[3]We retained duplicates of the same article from different media outlets to enable the measurement of media bubbles.

[4]fastText is an open-source library by Facebook AI Research for text classification and word representations; its pre-trained language identification model detects the language of a text with high accuracy. See https://fasttext.cc/.

[5]These metadata are very important: they enable modeling the probability of appearing on the front page and tracking coverage by individual journalists over time, among other analyses.

[6]Data extraction concluded in February 2025; consequently, we exclude 2025 from the yearly counts because the year is incomplete. We nevertheless plan to update the database regularly and create an observatory.

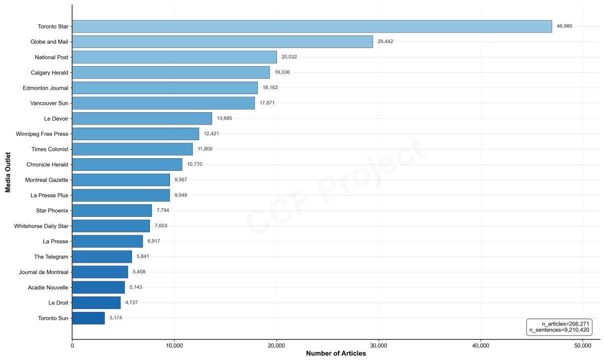

Figure 5: Distribution of articles by media.

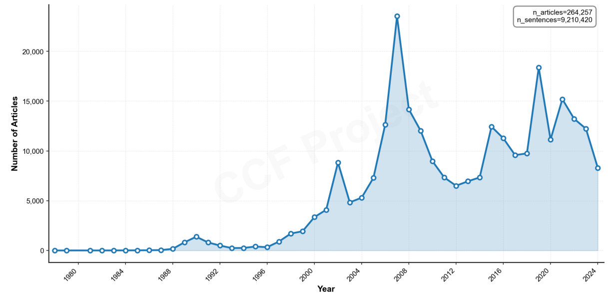

Figure 6: Total number of articles per year.

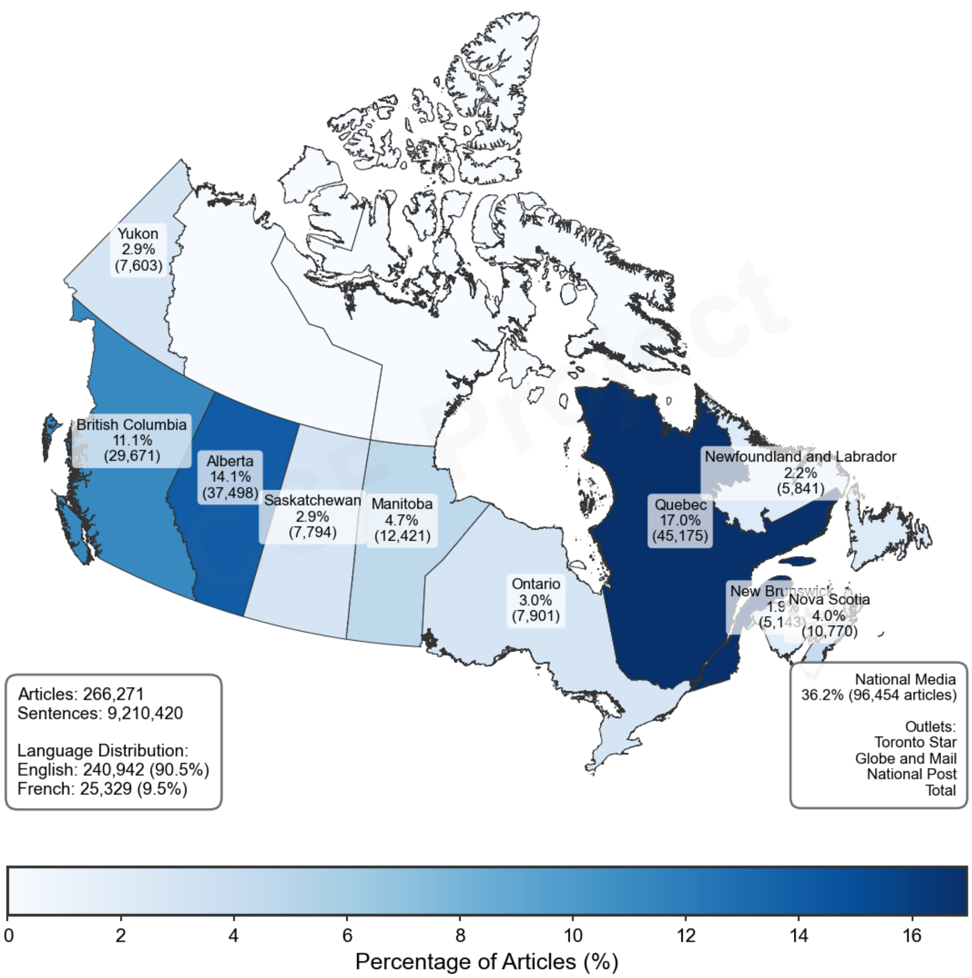

Figure 7: Distribution of articles by province.

entity-level extraction identifying and classifying named entities within the text. The complete detailed taxonomy with all 65 individual categories and their operational definitions is provided in Appendix **??** (Table B1), but in Table 2 we present a summary of the framework's structure and key categories.

The annotation framework possesses four primary categories (*Frames*, *Actors*, *Events* and *Solutions*) each with their own sub-categories. Each primary category is designed to detect the presence of the overarching dimension, while its sub-categories capture the internal distinctions within that dimension. For example, the *Economic Frame* is designed to categorize and detect any mention of economic aspects related to climate change, and it includes several categories that aim to capture specific economic sub-dimensions, such as negative/positive impacts of climate change on the economy, the costs and benefits of action, and economics sectors' footprint. The primary category *Actors/Messengers* captures any actor or messenger that is mentioned in the text and encompasses nine sub-categories, including scientists, politicians, activists, and various types of experts (medical, economic, security, legal, cultural); and so on for the primary categories.

The rest of the framework is composed of primary categories without sub-categories, including *Emotional Tone* (Positive, Negative, Neutral/none), *Geographic Focus* (Canadian location), *Urgency/Alarmism* (conveying immediate danger or crisis), and *Named Entities* (Persons, Organizations, Locations). These categories capture additional dimensions of climate discourse that are crucial for understanding the framing and context of climate change in Canadian media. The following real example from the database—an excerpt from a 2022 article—illustrates the corresponding manual annotations:

Example (annotated sentence).

"In Scientific American, professor of Environmental Studies at Humboldt State University Sarah Jaquette Ray writes, 'Climate anxiety can operate like white fragility, sucking up all the oxygen in the room and devoting resources toward appeasing the dominant group.' But it would be a mistake to conclude that this term isn't applicable in the Global South, period."

Dimension	**Tagged categories**
Actors	Scientists
Thematic frames	Justice (primary category) + North–South responsibility; Health (primary category)
Emotional tone	Negative emotion (anxiety)

A.3 Phase 3: Machine learning

A.3.1 Step 4 & 5: Iterative machine learning training and validation

The development of training data followed an iterative manual annotation protocol aligned with the annotation framework previously described. A single expert annotator (climate policy and communication) labeled sentences in sequential batches (first 2,000 sentences, then three additional batches of 1,000 each) for a total of 4,000 annotated samples. The batching served two explicit objectives: (1) to iteratively improve and stabilize the annotation

Table 2: The CCF annotation framework: 65 hierarchical categories for comprehensive climate discourse analysis

Main Category	N[a]	Sub-categories and Detection Elements
Primary Categories: Thematic Frames (8 frames with 35 sub-categories)		
Economic	6	Negative/positive impacts, costs/benefits of action, sector footprint, general link
Health	5	Negative/positive impacts, co-benefits, healthcare footprint, general link
Security	5	Military response, base disruption, displacement, conflicts, defense footprint
Justice	5	Winners/losers, North-South responsibility, legitimacy, litigation, general link
Political	3	Policy measures, political debate & opinion, general link
Scientific	3	Controversy, discovery & innovation, general link
Environmental	3	Habitat loss, species loss, general biodiversity link
Cultural	5	Art, events, Indigenous practices, cultural footprint, general link
Primary Categories: Actors, Events and Solutions (with 15 sub-categories)		
Actors/Messengers	9	Scientists, politicians, activists, medical/economic/security/legal/cultural experts, any messenger
Events	6	Natural disasters, conferences, reports, elections, policy announcements, any event
Solutions	3	Mitigation, adaptation, any solution mention
Emotional Tone	3	Positive (hope, optimism), Negative (fear, anxiety), Neutral/none
Geographic Focus	1	Canadian places, actors, data, and policies
Urgency/Alarmism	1	Conveys immediate danger, crisis, or "code red" urgency
Named Entities	3	Persons (PER), Organizations (ORG), Locations (LOC)
Total: 65 categories		

[a]The N count includes detection of the main category plus its sub-categories. For example: Economic frame detection + 5 sub-categories = 6.

guidelines after each round; and (2) to progressively monitor and optimize model training to reach the highest attainable F1 score. This process culminated in a macro F1 score of 0.826 (see Table 3) during the training phase (see Table B2 in Appendix **??** for detailed metrics).

The sampling process drew equally from English and French language groups, with final annotation counts reaching 1,927 English and 2,073 French sentences. The training dataset was subsequently partitioned using stratified random sampling to create training (80%) and validation (20%) databases for each category and language. Table B3 in Appendix **??** provides the complete distribution of training and validation samples for all annotation categories (**doAugmentedSocialScientist2022**). The random sampling approach ensured broad coverage across the temporal span and diverse media sources. While inter-annotator reliability assessment is in progress with a second coder, the annotation protocol was developed with detailed operational definitions for each of the 65 categories to ensure consistency.

The machine learning pipeline leveraged state-of-the-art transformer architectures optimized for each language through a custom fork of the AugmentedSocialScientist library (**doAugmentedSocialScientist2022**; **lemorAntoinelemorAugmentedSocialScientistFork2025**).

The fork we built from **doAugmentedSocialScientist2022** add several functionnalities that are central to ensure robust machine learning training. The fork provides: metric logging at every training epoch, intelligent best-model selection using a weighted F1 score (0.7 × F1-positive + 0.3 × macro-F1), automatic device optimization, and an automated reinforced training protocol for underperforming models (described below) (**lemorAntoinelemorAugmentedSocial** English text processing employed BERT-base-uncased, while French text utilized CamemBERT-base, both containing 110 million parameters. The training strategy implemented hierarchical classification, where detection models were trained first as binary classifiers on the full annotated dataset, followed by sub-classification models trained exclusively on manually annotated sentences that were positive for the corresponding primary category. This approach minimized false positive propagation and allowed for specialized optimization for each classification task.

Models failing to achieve positive-class F1 scores above 0.70 during the training phase underwent an automated reinforced training protocol[7]. The reinforced phase implemented weighted random sampling to oversample minority classes, doubled batch size to 64, reduced learning rate to 1e-5, and applied weighted cross-entropy loss with emphasis on the positive class. This reinforcement protocol, extending training for an additional 20 epochs, successfully improved mean F1 scores by 0.23 across affected models, and managed to bring most

[7]This protocol was necessary for 15 underperforming models, primarily in abstract conceptual categories such as cultural and justice sub-frames.

Table 3: Model training performance metrics for primary annotation categories

Category	F1 (Class 1)		F1 (Class 0)		Macro F1	
	EN	FR	EN	FR	EN	FR
Thematic Frames						
Economic Frame	0.745	0.814	0.944	0.957	0.845	0.885
Health Frame	0.800	0.667	0.989	0.994	0.894	0.830
Security Frame	0.870	0.800	0.996	0.997	0.933	0.898
Justice Frame	0.719	0.717	0.975	0.981	0.847	0.849
Political Frame	0.808	0.774	0.897	0.888	0.853	0.831
Scientific Frame	0.784	0.702	0.953	0.962	0.869	0.832
Environmental Frame	0.842	0.625	0.989	0.980	0.915	0.802
Cultural Frame	0.773	0.833	0.986	0.993	0.879	0.913
Other Primary Categories: Actors, Events and Solutions						
Actors/Messengers Detection	0.912	0.929	0.904	0.915	0.908	0.922
Event Detection	0.794	0.819	0.935	0.932	0.865	0.876
Solutions Detection	0.737	0.878	0.914	0.944	0.825	0.911
Canadian Context	0.942	0.964	0.968	0.980	0.955	0.972
Urgency/Alarmism	0.760	0.690	0.987	0.981	0.874	0.835
Overall Average*	**0.769**	**0.739**	**0.905**	**0.909**	**0.837**	**0.816**
Combined (ALL)	**0.754**		**0.907**		**0.826**	

*Overall average across all active annotation categories (see Table B2 for complete metrics).

categories above the acceptable performance threshold.

However, five categories could not be trained due to insufficient annotations in the training data and were consequently excluded from the annotation framework: carbon footprint of the health sector, positive health impacts of climate change, military disaster response, climate impacts on military operations, and carbon footprint of the security sector. These categories are not represented in the performance metrics table (Table 3).

A.4 Phase 4: Validation & Deployment

A.4.1 Step 6: Full Deployment

The full deployment phase applied all the trained models to annotate the complete CCF Database of 9.2 million sentences across 266,271 articles. We implemented a hierarchical classification strategy that directly mirrors the annotation protocol's structure of primary categories and their associated sub-categories, as illustrated in Figure 8.

The hierarchical approach reflects the conceptual organization of our annotation framework with eleven primary detection categories, each with specific sub-categories. The eight thematic frames (*Economic*, *Health*, *Security*, *Justice*, *Political*, *Scientific*, *Environmental*, and *Cultural Frame Detection*) function as primary categories with their internal distinctions—for example, *Economic Frame Detection*, when positive (=1), triggers six economic sub-models (*Negative/Positive Economic Impacts*, *Costs/Benefits of Climate Action*, *Economic Sector Footprint*), while a negative detection (=0) bypasses these entirely.

Similarly, the three other primary categories follow this conditional logic: *Actors/Messengers Detection* activates nine actor-type sub-models (*Health Expert*, *Economic Expert*, *Security*

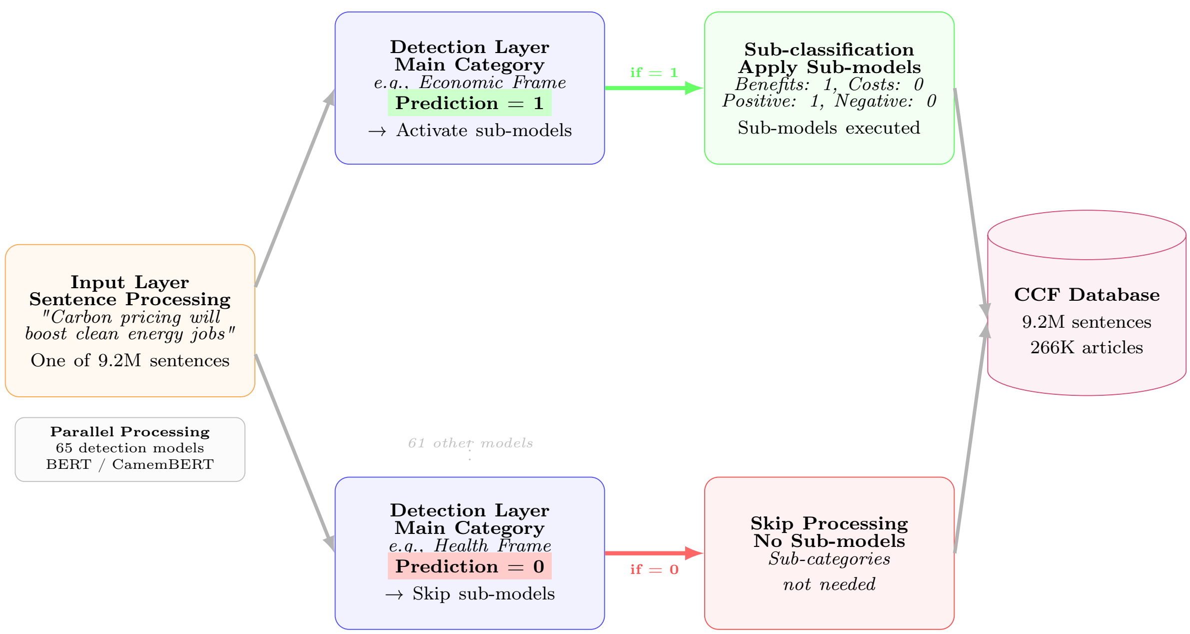

Figure 8: Hierarchical annotation pipeline. The system processes each sentence through detection models that determine whether corresponding sub-category models should be applied, enabling efficient annotation of the entire corpus.

Expert, *Legal Expert*, *Cultural Expert*, *Natural Scientist*, *Social Scientist*, *Activist*, *Public Official*), *Event Detection* triggers eight event-type classifications (*Extreme Weather Event*, *Meeting/Conference*, *Publication*, *Election*, *Policy Announcement*, *Judiciary Decision*, *Cultural Event*, *Protest*), and *Solutions Detection* applies two solution sub-categories (*Mitigation Strategy*, *Adaptation Strategy*). The deployment also processed standalone primary categories without hierarchical structure—*Emotional Tone* (*Positive Emotion*/*Negative Emotion*/*Neutral Emotion*), *Geographic Focus* (*Canadian Context*), *Urgency/Alarmism*.

In addition to the custom-trained classification models, we employed pre-trained Named Entity Recognition (NER) models, selected based on empirical evaluation for optimal performance on our dataset. We implemented a hybrid approach tailored to each language: for English texts, we used BERT-base-NER[8] for all three entity types (Person, Organization, Location), while for French texts, we combined spaCy's fr_core_news_lg model[9] for person entities with CamemBERT-NER[10] for organization and location entities. This hybrid strategy was used to leverage the respective strengths of each model: spaCy's superior performance on French person names and CamemBERT-NER's robust identification of French organizational and geographical entities (see Table B4 in the Appendix for performance metrics).

The resulting CCF database structure, illustrated in Figure 9, organizes the annotations in a relational PostgreSQL schema. Each sentence in the database maintains its connection to the source article metadata (publication, date, author) while storing the 65 annotation predictions as indexed boolean columns (or JSON for NER entities). The database design

[8] Available at: https://huggingface.co/dslim/bert-base-NER
[9] Available at: https://spacy.io/models/fr
[10] Available at: https://huggingface.co/Jean-Baptiste/camembert-ner

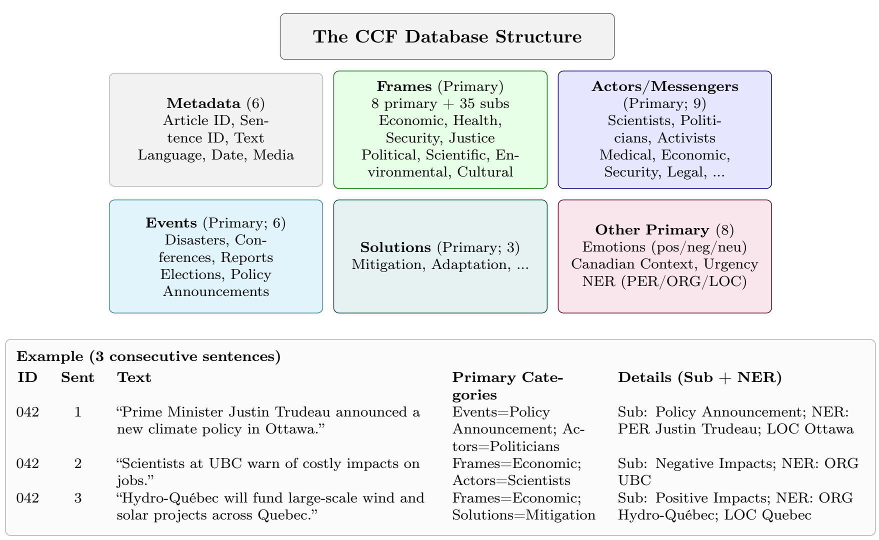

Figure 9: The CCF database structure (synthesized view) stores 9.2 million sentences and 71 boolean columns grouped into six families: Metadata, Frames (8 frames with 35 subcategories), Actors/Messengers (with its subcategories), Events (with its subcategories), Solutions (with its subcategories), and other primary categories (Emotional tone, Geographic Focus, Urgence/alarmism, NER).

supports complex multi-dimensional analyses. Researchers can, for instance, query all sentences between 2015-2020 that contain economic frames with positive emotions and mention specific political actors (or even specific named entities such as Justin Trudeau).

A.4.2 Step 7 & 8: Final validation and inter-coder reliability

The final validation phase employs a stratified sampling scheme designed to ensure robust performance evaluation across all 65 annotation categories. Following the completion of full database annotation (9.2 million sentences), we extracted a balanced validation sample of 1,000 sentences (500 per language) using root-inverse probability weighting[11] combined with strict constraints to ensure statistical validity. Two critical constraints were used : first, a minimum threshold of 40 positive examples per category to ensure sufficient statistical power for reliable performance estimation. Without this constraint, rare categories like *Military Base Disruption* (0.00004% prevalence in the database) would have too few examples to meaningfully assess model accuracy. Second, a 35% maximum allocation to prevent any single prevalent category from monopolizing the validation set. For instance, without this constraint, *Actors/Messengers Detection* (present in 48.8% of database sentences) could theoretically dominate the entire sample, leaving insufficient space for other important categories.

The validation results demonstrate robust performance across all annotation dimensions, with an overall F1 macro score of 0.866 (0.869 for English, 0.864 for French). As detailed in Table B5 (Appendix), primary detection categories achieve the highest performance, with *Canadian Context* (F1=0.982) and *Actors/Messengers Detection* (F1=0.961) showing near-perfect classification. Thematic frames maintain strong performance despite their semantic complexity, with *Scientific Frame Detection* (F1=0.890) and *Environmental Frame Detection* (F1=0.871) leading this category. The hierarchical classification strategy proves particularly

[11]The weighting formula $w_i \propto \sum_j 1/\sqrt{f_j}$ assigns higher sampling probability to sentences containing rare categories, where f_j represents the frequency of category j present in sentence i. For example, a sentence annotated with both *Economic Frame* (appearing in 15.4% of the corpus) and *Loss of Indigenous Practices* (appearing in 0.28% of the corpus) would receive a weight of approximately $1/\sqrt{0.154}+1/\sqrt{0.0028} = 2.55 + 18.90 = 21.45$, making it roughly 8 times more likely to be selected than a sentence containing only *Economic Frame*.

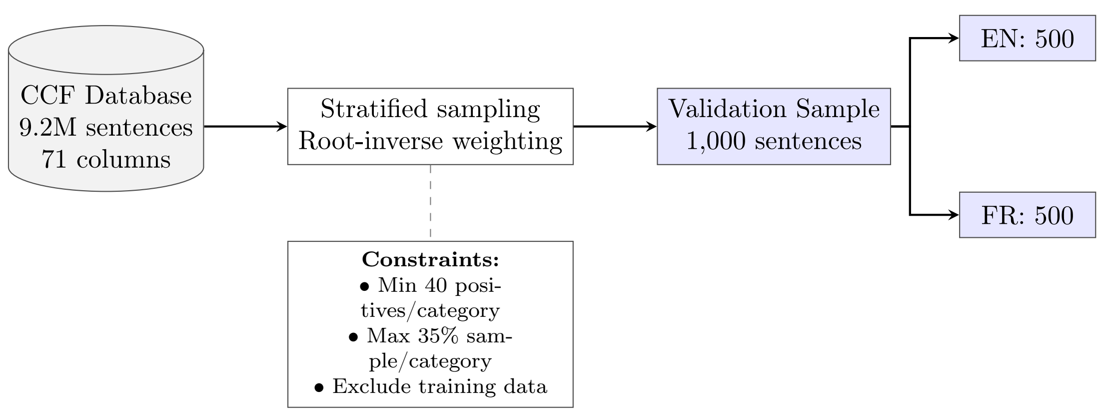

Figure 10: Stratified sampling procedure for final validation. Root-inverse probability weighting ensures balanced representation across all 65 annotation categories.

effective, as evidenced by consistently higher recall scores for detection categories (mean recall=0.891) that trigger conditional sub-category evaluation. Notably, rare categories such as *Military Base Disruption* and *Military Disaster Response* show perfect precision despite minimal representation.

To assess temporal stability, we conducted stratified validation across five temporal periods (1980s-2020s) that revealed minimal performance drift (ΔF1 $<$ 1.1%) despite evolving media discourse patterns (Figure 11). This temporal consistency validates the models' generalization capacity across the full 46-year corpus span. Inter-coder reliability assessment, currently in progress with a second independent annotator, will evaluate the same 1000 randomly selected sentences from the validation set. Final inter-coder metrics will be reported upon completion of the dual-annotation process.

Table 4: Overall validation performance metrics

Language	**F1 Macro**	**F1 Micro**	**F1 Weighted**
English (EN)	0.869	0.909	0.911
French (FR)	0.864	0.902	0.905
Combined (ALL)	**0.866**	**0.905**	**0.908**

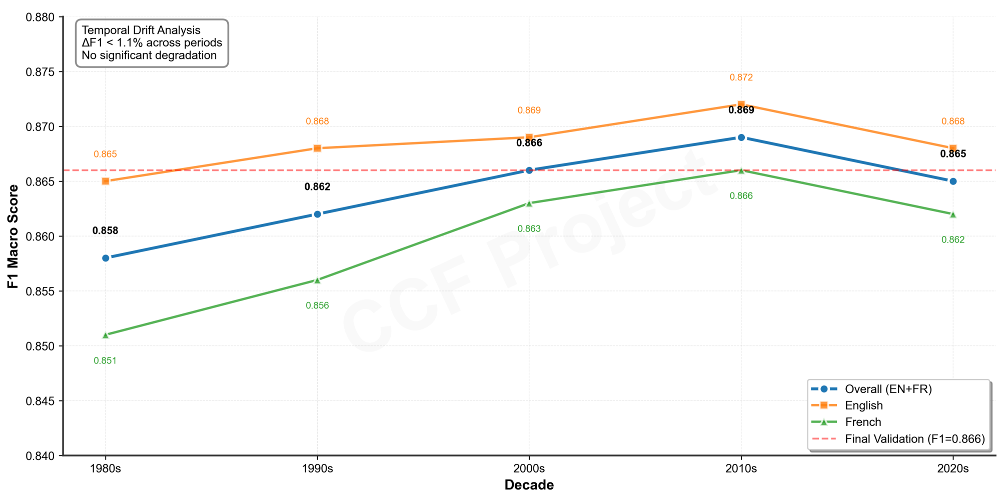

Figure 11: Temporal validation showing stable F1 macro scores across five time periods. The minimal variation (ΔF1 $<$ 1.1%) confirms robust model performance without temporal degradation.

B Complete Annotation Framework and Performance Tables

Table B1: Complete CCF annotation framework: All 65 categories with operational definitions

#	Category	Description
THEMATIC FRAMES		
Economic Frame		
1	Economic Frame Detection	Presence of any economic dimension of climate change
2	Negative Economic Impacts	Economic losses from climate change (e.g., floods, infrastructure damage)
3	Positive Economic Impacts	Economic benefits from climate action (e.g., green jobs, energy savings)
4	Costs of Climate Action	Economic costs of transitioning to low-carbon economy
5	Benefits of Climate Action	Economic opportunities from climate investments
6	Economic Sector Footprint	Carbon intensity of economic sectors
Health Frame		
7	Health Frame Detection	Presence of any relationship between climate and health
8	Negative Health Impacts	Heat stress, disease spread, respiratory issues, mental health burdens, mortality
9	Health Co-benefits	Better air quality, improved diets, avoided premature deaths, mental well-being
Security Frame		
10	Security Frame Detection	Presence of any security dimension
11	Climate-Driven Displacement	Military management of evacuations or refugee camps
12	Resource Conflict	Tensions or violence over water, land or minerals worsened by climate change
13	*Military Disaster Response**	Army called in for fires, floods, evacuations or relief
14	*Military Base Disruption**	Climate threats to military facilities or readiness
Justice Frame		
15	Justice Frame Detection	Presence of any social justice angle
16	Winners & Losers	Groups that benefit or suffer from climate measures (workers, vulnerable populations)
17	North-South Responsibility	Common-but-differentiated responsibilities between high and low-income countries
18	Unequal Impacts	Disproportionate climate impacts on vulnerable groups
19	Unequal Access	Differential access to resources, adaptation, or clean technology
20	Intergenerational Justice	Responsibilities to future generations
Political Frame		
21	Political Frame Detection	Presence of any policy measure or political discussion

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Table B1 – *Continued from previous page*

#	Category	Description
22	Policy Measures	Concrete climate laws, regulations, or programmes under debate or in force
23	Political Debate	Parliamentary disputes, party platforms
24	Political Positioning	Strategic positioning or partisan signalling around climate
25	Public Opinion	Polls or surveys on climate and energy
Scientific Frame		
26	Scientific Frame Detection	Presence of any scientific aspect
27	Scientific Controversy	Debates on climate change reality, causes, thresholds, geoengineering ethics
28	Discovery & Innovation	New findings on climate impacts or emerging technologies (e.g., carbon capture)
29	Scientific Uncertainty	Expressions of doubt or uncertainty about climate science
30	Scientific Certainty	Strong consensus statements about climate science
Environmental Frame		
31	Environmental Frame Detection	Presence of any biodiversity concern
32	Habitat Loss	Glacier melt, coral bleaching, forest die-off, wetland drying
33	Species Loss	Local or global extinction risk for animals or plants
Cultural Frame		
34	Cultural Frame Detection	Presence of any cultural aspect
35	Artistic Representation	Books, documentaries, plays, exhibitions portraying climate themes
36	Event Disruption	Sports or cultural events threatened or cancelled due to climate conditions
37	Loss of Indigenous Practices	Erosion of traditional hunting, fishing, or cultural rituals linked to climate
38	Cultural Sector Footprint	Emissions from film production, fashion, large festivals, etc.
PRIMARY CATEGORIES		
Actors/Messengers		
39	Actors/Messengers Detection	Presence of any messenger, expert or authority figure
40	Health Expert	Physicians, epidemiologists, health ministers, public health officials
41	Economic Expert	Economists, finance ministers, market analysts, central bank officials
42	Security Expert	Military officers, defense strategists, security scholars
43	Legal Expert	Lawyers, judges, legal scholars, justice ministers
44	Cultural Expert	Artists, writers, athletes, arts scholars commenting on climate change
45	Natural Scientist	Natural science researchers or academics
46	Social Scientist	Social science researchers or academics
47	Activist	Environmental NGO spokespeople or well-known climate activists
48	Public Official	Civil servants or government officials speaking in an official capacity

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Table B1 – *Continued from previous page*

#	Category	Description
Events		
49	Event Detection	Mentions at least one specific event type
50	Extreme Weather Event	Arrival or unfolding of floods, wildfires, hurricanes, heatwaves, etc.
51	Meeting/Conference	International meetings such as COP, UN summits, major national conferences
52	Publication	Publication of governmental, NGO or scientific reports (e.g., IPCC, Lancet)
53	Election	Climate issues raised during local, provincial or national elections
54	Policy Announcement	Debut or unveiling of new climate laws, regulations or action plans
55	Judiciary Decision	Court rulings, trials, regulatory hearings on climate or environment
56	Cultural Event	Sports, artistic, or cultural events (Olympics, marathon, concert, theatre)
57	Protest	Demonstrations or strikes (e.g., climate strike, anti-pipeline protest)
Solutions		
58	Solutions Detection	Mentions any mitigation or adaptation measure
59	Mitigation Strategy	Measures to reduce GHG emissions or enhance carbon sinks
60	Adaptation Strategy	Measures to increase social or ecological resilience to climate impacts
EMOTIONAL TONE		
61	Positive Emotion	Hope, optimism, enthusiasm about climate solutions
62	Negative Emotion	Fear, anxiety, despair about climate impacts
63	Neutral Emotion	No clear emotional tone, factual reporting
GEOGRAPHIC FOCUS		
64	Canadian Context	Canadian places, actors, data, and policies
URGENCY/ALARMISM		
65	Urgency/Alarmism	Conveys immediate danger, crisis, or “code red” urgency

* Insufficient training data (fewer than 10 positive examples in training set).

Table B2: Complete model training performance metrics for all annotation categories

#	Category	F1 (Class 1) EN	F1 (Class 1) FR	F1 (Class 0) EN	F1 (Class 0) FR	Macro F1 EN	Macro F1 FR
THEMATIC FRAMES							
Economic Frame							
1	Economic Frame Detection	0.745	0.814	0.944	0.957	0.845	0.885
2	Negative Economic Impacts	0.727	0.889	0.914	0.944	0.821	0.917
3	Positive Economic Impacts	0.333	0.400	0.995	0.996	0.664	0.698
4	Costs of Climate Action	0.615	0.500	0.848	0.913	0.732	0.707
5	Benefits of Climate Action	0.500	0.516	0.952	0.981	0.726	0.749
6	Economic Sector Footprint	0.857	0.857	0.938	0.950	0.897	0.904
Health Frame							
7	Health Frame Detection	0.800	0.667	0.989	0.994	0.894	0.830
8	Negative Health Impacts	0.909	0.857	0.000	0.000	0.455	0.429
9	Positive Health Impacts	–	–	–	–	–	–
10	Health Co-benefits	0.400	0.011	0.571	0.227	0.486	0.119
11	Health Sector Footprint	–	–	–	–	–	–
Security Frame							
12	Security Frame Detection	0.870	0.898	0.996	0.997	0.933	0.898
13	Military Disaster Response	–	0.026	–	1.000	–	0.026
14	Military Base Disruption	–	1.000	–	1.000	–	1.000
15	Climate-Driven Displacement	1.000	1.000	1.000	1.000	1.000	1.000
16	Resource Conflict	1.000	1.000	1.000	1.000	1.000	1.000
17	Defense Sector Footprint	–	–	–	–	–	–
Justice Frame							
18	Justice Frame Detection	0.719	0.717	0.975	0.981	0.847	0.849
19	Winners & Losers	0.667	0.667	0.933	0.923	0.800	0.795
20	North-South Responsibility	0.857	0.750	0.909	0.750	0.883	0.750
21	Unequal Impacts	0.571	0.600	0.992	0.995	0.782	0.798
22	Unequal Access	0.364	0.625	0.000	0.993	0.182	0.809
23	Intergenerational Justice	0.933	0.800	0.999	0.999	0.966	0.899
Political Frame							
24	Political Frame Detection	0.808	0.774	0.897	0.888	0.853	0.831
25	Policy Measures	0.621	0.667	0.985	0.966	0.803	0.816
26	Political Debate	0.916	0.966	0.222	0.571	0.569	0.768
27	Political Positioning	0.750	0.800	0.946	0.989	0.848	0.895
28	Public Opinion	0.909	1.000	0.999	1.000	0.954	1.000
Scientific Frame							
29	Scientific Frame Detection	0.784	0.702	0.953	0.962	0.869	0.832
30	Scientific Controversy	0.737	0.500	0.848	0.875	0.793	0.688
31	Discovery & Innovation	0.828	0.938	0.800	0.750	0.814	0.844
32	Scientific Uncertainty	0.727	0.333	0.927	0.995	0.827	0.664
33	Scientific Certainty	0.571	0.000	0.933	0.919	0.752	0.459
Environmental Frame							
34	Environmental Frame Detection	0.842	0.625	0.989	0.980	0.915	0.802
35	Habitat Loss	1.000	0.857	1.000	0.800	1.000	0.829
36	Species Loss	0.889	0.857	0.857	0.800	0.873	0.829
Cultural Frame							
37	Cultural Frame Detection	0.773	0.833	0.986	0.993	0.879	0.913

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Table B2 – *Continued from previous page*

#	Category	F1 (Class 1)		F1 (Class 0)		Macro F1	
		EN	**FR**	**EN**	**FR**	**EN**	**FR**
38	Artistic Representation	1.000	1.000	1.000	1.000	1.000	1.000
39	Event Disruption	0.706	1.000	0.993	1.000	0.850	1.000
40	Loss of Indigenous Practices	1.000	0.026	1.000	0.899	1.000	0.462
41	Cultural Sector Footprint	1.000	0.005	1.000	0.143	1.000	0.074
PRIMARY CATEGORIES							
Actors/Messengers							
42	Actors/Messengers Detection	0.912	0.929	0.904	0.915	0.908	0.922
43	Health Expert	0.857	0.909	0.997	0.999	0.927	0.954
44	Economic Expert	0.600	0.750	0.970	0.973	0.785	0.861
45	Security Expert	0.667	1.000	0.993	1.000	0.830	1.000
46	Legal Expert	0.667	0.769	0.999	0.996	0.833	0.883
47	Cultural Expert	1.000	0.556	1.000	0.990	1.000	0.773
48	Natural Scientist	0.789	0.833	0.977	0.971	0.883	0.902
49	Social Scientist	0.769	0.645	0.996	0.986	0.883	0.816
50	Activist	0.571	1.000	0.978	1.000	0.775	1.000
51	Public Official	0.703	0.923	0.897	0.964	0.800	0.943
Events							
52	Event Detection	0.794	0.819	0.935	0.932	0.865	0.876
53	Extreme Weather Event	1.000	0.857	1.000	0.968	1.000	0.912
54	Meeting/Conference	0.824	0.957	0.936	0.982	0.880	0.969
55	Publication	0.873	1.000	0.990	1.000	0.931	1.000
56	Election	1.000	0.800	1.000	0.998	1.000	0.899
57	Policy Announcement	0.769	0.727	0.943	0.954	0.856	0.841
58	Judiciary Decision	1.000	1.000	1.000	1.000	1.000	1.000
59	Cultural Event	0.333	1.000	0.995	1.000	0.664	1.000
60	Protest	0.889	0.727	0.999	0.996	0.944	0.862
Solutions							
61	Solutions Detection	0.737	0.878	0.914	0.944	0.825	0.911
62	Mitigation Strategy	0.750	0.812	0.935	0.942	0.842	0.877
63	Adaptation Strategy	0.696	0.800	0.991	0.989	0.843	0.894
EMOTIONAL TONE							
62	Positive Emotion	0.526	0.690	0.966	0.968	0.746	0.829
63	Negative Emotion	0.706	0.765	0.785	0.843	0.745	0.804
64	Neutral Emotion	0.741	0.789	0.676	0.693	0.709	0.741
GEOGRAPHIC FOCUS							
65	Canadian Context	0.942	0.964	0.968	0.980	0.955	0.972
URGENCY/ALARMISM							
66	Urgency/Alarmism	0.591	0.649	0.975	0.984	0.783	0.816
AVERAGE PERFORMANCE METRICS							
	English Average	0.769	–	0.905	–	0.837	–
	French Average	–	0.739	–	0.909	–	0.816
	Overall Average	0.754		0.907		0.826	
TOTAL: 65 ANNOTATION CATEGORIES *(62 categories with at least one model; 3 categories entirely excluded*)*							

* Insufficient training data (fewer than 10 positive examples in training set).

Table B3: Training and validation dataset distribution across all annotation categories

#	Category	English				French			
		Training		Validation		Training		Validation	
		Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
THEMATIC FRAMES									
Economic Frame									
1	Economic Frame Detection	218	1067	24	118	252	1164	28	129
2	Negative Economic Impacts	61	158	6	17	75	178	8	19
3	Positive Economic Impacts	14	205	1	22	7	245	1	27
4	Costs of Climate Action	63	156	6	17	42	211	4	23
5	Benefits of Climate Action	36	183	3	20	48	205	5	22
6	Economic Sector Footprint	67	152	7	16	77	176	8	19
Health Frame									
7	Health Frame Detection	57	1228	6	136	37	1379	4	153
8	Negative Health Impacts	47	10	5	1	32	5	3	1
9	Positive Health Impacts	0	56	1	6	0	36	1	4
10	Health Co-benefits	8	49	1	5	4	33	1	3
11	Health Sector Footprint	0	56	1	6	0	37	0	4
Security Frame									
12	Security Frame Detection	19	1266	2	140	19	1397	2	155
13	Military Disaster Response	0	18	1	2	4	15	1	1
14	Military Base Disruption	0	18	1	2	1	18	1	1
15	Climate-Driven Displacement	10	9	1	1	7	12	1	1
16	Resource Conflict	6	13	1	1	7	12	1	1
17	Defense Sector Footprint	0	19	0	2	0	19	0	2
Justice Frame									
18	Justice Frame Detection	90	1196	9	132	81	1335	9	148
19	Winners & Losers	20	70	2	7	20	62	2	6
20	North-South Responsibility	38	52	4	5	41	41	4	4
21	Unequal Impacts	8	81	1	9	9	72	1	8
22	Unequal Access	21	69	2	7	20	62	2	6
23	Intergenerational Justice	14	76	1	8	6	75	1	8
Political Frame									
24	Political Frame Detection	416	869	46	96	440	977	48	108
25	Policy Measures	62	355	6	39	58	382	6	42
26	Political Debate	344	72	38	8	396	45	43	4
27	Political Positioning	61	356	6	39	35	405	3	45
28	Public Opinion	21	396	2	43	20	420	2	46
Scientific Frame									
29	Scientific Frame Detection	243	1042	27	115	185	1232	20	136
30	Scientific Controversy	98	146	10	16	46	139	5	15
31	Discovery & Innovation	126	117	14	13	144	41	16	4
32	Scientific Uncertainty	50	194	5	21	17	169	1	18
33	Scientific Certainty	37	207	4	22	32	153	3	17
Environmental Frame									
34	Environmental Frame Detection	84	1201	9	133	63	1354	6	150
35	Habitat Loss	50	35	5	3	35	28	3	3
36	Species Loss	42	43	4	4	32	31	3	3
Cultural Frame									

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Table B3 – *Continued from previous page*

#	Category	English				French			
		Training		Validation		Training		Validation	
		Pos	Neg	Pos	Neg	Pos	Neg	Pos	Neg
37	Cultural Frame Detection	44	1242	4	137	45	1371	5	152
38	Artistic Representation	20	24	2	2	18	27	2	3
39	Event Disruption	11	33	1	3	21	25	2	2
40	Loss of Indigenous Practices	10	34	1	3	4	41	1	4
41	Cultural Sector Footprint	1	42	1	4	2	43	1	4
PRIMARY CATEGORIES									
Actors/Messengers									
42	Actors/Messengers Detection	657	629	72	69	736	681	81	75
43	Health Expert	10	647	1	71	8	728	1	80
44	Economic Expert	55	602	6	66	67	669	7	74
45	Security Expert	3	653	1	72	10	726	1	80
46	Legal Expert	1	655	1	72	4	731	1	81
47	Cultural Expert	13	644	1	71	9	727	1	80
48	Natural Scientist	107	550	11	61	117	620	12	68
49	Social Scientist	9	648	1	71	26	711	2	78
50	Activist	53	604	5	67	55	681	6	75
51	Public Official	171	486	18	54	223	513	24	57
Events									
52	Event Detection	298	987	33	109	355	1062	39	117
53	Extreme Weather Event	72	226	8	25	59	297	6	32
54	Meeting/Conference	75	224	8	24	100	255	11	28
55	Publication	74	225	8	24	114	242	12	26
56	Election	16	283	1	31	18	337	2	37
57	Policy Announcement	55	243	6	27	60	296	6	32
58	Judiciary Decision	10	288	1	32	10	345	1	38
59	Cultural Event	2	296	1	32	11	344	1	38
60	Protest	11	288	1	31	8	347	1	38
Solutions									
61	Solutions Detection	314	972	34	107	415	1001	46	111
EMOTIONAL TONE									
62	Positive Emotion	103	1182	11	131	151	1266	16	140
63	Negative Emotion	491	794	54	88	537	880	59	97
64	Neutral Emotion	686	599	76	66	726	691	80	76
GEOGRAPHIC FOCUS									
65	Canadian Context	445	840	49	93	493	924	54	102
URGENCY/ALARMISM									
66	Urgency/Alarmism	66	1219	7	135	63	1354	6	150

Note: Categories with insufficient positive training samples (marked with * in Table B2) were excluded from final model training and are not shown in this distribution table.

Table B4: Named Entity Recognition model performance metrics

Language	Model	PER F1	ORG F1	LOC F1
English	BERT-base-NER	0.961	0.811	0.925
French	spaCy fr_core_news_lg	0.880	–	–
	CamemBERT-NER	–	0.824	0.929

Note: The hybrid approach for French combines spaCy for person entities (PER) with CamemBERT-NER for organization (ORG) and location (LOC) entities based on empirical evaluation on our dataset. Performance metrics are from the original model documentation.

Table B5: Detailed validation performance metrics for trained models

#	Category	F1 Macro			F1 Micro			F1 Weighted			Support		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
THEMATIC FRAMES													
Economic Frame													
1	Economic Frame Detection	0.847	0.814	0.830	0.892	0.864	0.878	0.890	0.862	0.875	120	129	249
2	Negative Economic Impacts	0.820	0.741	0.783	0.848	0.810	0.828	0.856	0.833	0.842	26	18	44
3	Positive Economic Impacts	0.806	0.508	0.684	0.905	0.845	0.873	0.913	0.908	0.900	12	1	13
4	Costs of Climate Action	0.741	0.702	0.723	0.800	0.802	0.801	0.813	0.806	0.810	22	23	45
5	Benefits of Climate Action	0.885	0.817	0.848	0.924	0.871	0.896	0.928	0.872	0.899	19	26	45
6	Economic Sector Footprint	0.785	0.788	0.787	0.838	0.845	0.842	0.831	0.843	0.837	30	29	59
Health Frame													
7	Health Frame Detection	0.800	0.817	0.808	0.947	0.961	0.954	0.947	0.964	0.956	35	23	58
8	Negative Health Impacts	0.386	0.340	0.364	0.629	0.514	0.571	0.485	0.349	0.416	22	18	40
9	Health Co-benefits	0.252	0.103	0.186	0.257	0.114	0.186	0.301	0.023	0.177	4	4	8
Security Frame													
10	Security Frame Detection	0.825	0.730	0.782	0.951	0.939	0.945	0.955	0.950	0.952	30	19	49
11	Climate-Driven Displacement	0.476	0.417	0.481	0.545	0.429	0.488	0.468	0.476	0.459	21	6	27
12	Resource Conflict	0.358	0.286	0.324	0.364	0.286	0.326	0.398	0.286	0.346	7	6	13
13	Military Disaster Response	–	0.045	0.045	–	0.048	0.048	–	0.004	0.004	–	2	2
14	Military Base Disruption	–	1.000	1.000	–	1.000	1.000	–	1.000	1.000	–	2	2
Justice Frame													
15	Justice Frame Detection	0.832	0.880	0.857	0.917	0.937	0.927	0.921	0.939	0.930	62	74	136
16	Winners & Losers	0.654	0.737	0.708	0.765	0.762	0.764	0.802	0.775	0.786	10	22	32
17	North-South Responsibility	0.900	0.823	0.860	0.926	0.833	0.879	0.927	0.838	0.883	19	27	46
18	Unequal Impacts	0.786	0.894	0.845	0.852	0.917	0.885	0.858	0.918	0.888	16	21	37
19	Unequal Access	0.475	0.796	0.651	0.481	0.833	0.661	0.456	0.835	0.674	27	23	50
20	Intergenerational Justice	0.924	0.952	0.942	0.963	0.964	0.964	0.965	0.965	0.965	10	19	29
Political Frame													
21	Political Frame Detection	0.812	0.842	0.829	0.827	0.844	0.836	0.829	0.845	0.837	167	217	384
22	Policy Measures	0.802	0.722	0.756	0.914	0.864	0.886	0.909	0.854	0.879	26	38	64
23	Political Debate	0.469	0.650	0.564	0.694	0.820	0.763	0.595	0.781	0.699	127	175	302

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Table B5 – *Continued from previous page*

#	Category	F1 Macro			F1 Micro			F1 Weighted			Support		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
24	Political Positioning	0.752	0.808	0.782	0.801	0.912	0.862	0.821	0.920	0.877	35	24	59
25	Public Opinion	0.877	0.910	0.894	0.962	0.974	0.969	0.966	0.976	0.971	12	15	27
Scientific Frame													
26	Scientific Frame Detection	0.891	0.890	0.890	0.951	0.949	0.950	0.948	0.947	0.947	74	75	149
27	Scientific Controversy	0.962	0.867	0.911	0.962	0.869	0.912	0.962	0.869	0.912	26	26	52
28	Discovery & Innovation	0.884	0.797	0.838	0.885	0.803	0.841	0.886	0.803	0.842	31	36	67
29	Scientific Uncertainty	0.783	0.768	0.775	0.846	0.836	0.841	0.860	0.857	0.858	9	9	18
30	Scientific Certainty	0.769	0.460	0.682	0.846	0.852	0.850	0.862	0.785	0.843	8	9	17
Environmental Frame													
31	Environmental Frame Detection	0.866	0.875	0.871	0.967	0.967	0.967	0.969	0.968	0.969	28	32	60
32	Habitat Loss	0.804	0.573	0.721	0.806	0.707	0.753	0.802	0.637	0.730	16	26	42
33	Species Loss	0.714	0.692	0.703	0.722	0.707	0.714	0.717	0.694	0.705	19	21	40
Cultural Frame													
34	Cultural Frame Detection	0.802	0.581	0.708	0.923	0.880	0.901	0.931	0.920	0.921	41	11	52
35	Artistic Representation	0.834	0.469	0.659	0.851	0.559	0.704	0.858	0.649	0.739	18	6	24
36	Event Disruption	0.631	0.354	0.484	0.731	0.456	0.593	0.777	0.595	0.686	8	2	10
37	Loss of Indigenous Practices	0.824	0.415	0.595	0.910	0.618	0.763	0.922	0.749	0.825	7	1	8
38	Cultural Sector Footprint	0.702	0.014	0.351	0.925	0.015	0.467	0.945	0.014	0.613	2	1	3
PRIMARY CATEGORIES													
Actors/Messengers													
39	Actors/Messengers Detection	0.955	0.967	0.961	0.957	0.969	0.963	0.957	0.969	0.963	301	312	613
40	Health Expert	0.862	0.951	0.907	0.964	0.987	0.975	0.966	0.987	0.976	19	21	40
41	Economic Expert	0.847	0.895	0.874	0.954	0.961	0.957	0.949	0.963	0.957	30	28	58
42	Security Expert	0.728	0.813	0.787	0.970	0.958	0.964	0.973	0.962	0.968	7	14	21
43	Legal Expert	0.861	0.927	0.914	0.990	0.980	0.985	0.989	0.982	0.986	7	19	26
44	Cultural Expert	0.853	0.825	0.843	0.954	0.967	0.961	0.954	0.969	0.961	26	13	39
45	Natural Scientist	0.920	0.896	0.907	0.960	0.941	0.951	0.960	0.944	0.952	44	46	90
46	Social Scientist	0.938	0.948	0.944	0.987	0.984	0.985	0.987	0.984	0.986	17	24	41
47	Activist	0.934	0.903	0.918	0.977	0.964	0.970	0.976	0.963	0.970	31	33	64
48	Public Official	0.927	0.930	0.929	0.934	0.935	0.934	0.934	0.935	0.934	104	113	217
Events													

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Table B5 – *Continued from previous page*

#	Category	F1 Macro			F1 Micro			F1 Weighted			Support		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**	**EN**	**FR**	**ALL**
49	Event Detection	0.881	0.912	0.897	0.886	0.913	0.900	0.885	0.913	0.900	201	222	423
50	Extreme Weather Event	0.887	0.891	0.889	0.935	0.955	0.946	0.938	0.958	0.949	29	21	50
51	Meeting/Conference	0.855	0.891	0.877	0.908	0.914	0.911	0.914	0.918	0.916	30	50	80
52	Publication	0.950	0.962	0.958	0.973	0.968	0.970	0.973	0.969	0.971	30	62	92
53	Election	0.911	0.895	0.903	0.962	0.964	0.963	0.960	0.963	0.962	25	22	47
54	Policy Announcement	0.821	0.837	0.829	0.886	0.914	0.901	0.895	0.915	0.906	29	33	62
55	Judiciary Decision	0.628	0.939	0.808	0.903	0.982	0.946	0.927	0.981	0.951	6	19	25
56	Cultural Event	0.906	0.857	0.881	0.973	0.964	0.968	0.971	0.967	0.968	17	12	29
57	Protest	0.982	0.909	0.943	0.995	0.973	0.983	0.995	0.973	0.983	16	18	34
Solutions													
58	Solutions Detection	0.891	0.909	0.900	0.921	0.931	0.926	0.922	0.930	0.926	114	133	247
59	Mitigation Strategy	0.872	0.801	0.839	0.884	0.839	0.861	0.882	0.850	0.864	76	97	173
60	Adaptation Strategy	0.889	0.903	0.895	0.934	0.952	0.943	0.934	0.954	0.944	22	16	38
EMOTIONAL TONE													
61	Positive Emotion	0.712	0.692	0.702	0.902	0.892	0.897	0.910	0.904	0.907	38	37	75
62	Negative Emotion	0.793	0.804	0.799	0.807	0.813	0.810	0.814	0.819	0.816	151	165	316
63	Neutral Emotion	0.756	0.798	0.777	0.764	0.809	0.787	0.765	0.807	0.787	295	306	601
GEOGRAPHIC FOCUS													
64	Canadian Context	0.986	0.978	0.982	0.986	0.978	0.982	0.986	0.978	0.982	235	255	490
URGENCY/ALARMISM													
65	Urgency/Alarmism	0.874	0.835	0.853	0.976	0.965	0.970	0.977	0.967	0.972	23	25	48
OVERALL PERFORMANCE													
	All Categories	**0.869**	**0.864**	**0.866**	**0.909**	**0.902**	**0.905**	**0.911**	**0.905**	**0.908**	**3,069**	**3,332**	**6,401**

Note: Metrics represent macro-averaged scores across positive and negative classes for each category. Support indicates the number of positive examples in the validation set (500 sentences per language, 1,000 total). EN = English, FR = French, ALL = Combined. Categories are organized by their hierarchical grouping as implemented in the annotation pipeline.

Table B6: Database-wide distribution of annotation categories across 9.2 million sentences

#	Category	Count			Proportion (%)		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**
THEMATIC FRAMES							
Economic Frame							
1	Economic Frame Detection	1,235,686	185,032	1,420,718	15.95	12.64	15.43
2	Costs of Climate Action	517,576	40,259	557,835	6.68	2.75	6.06
3	Economic Sector Footprint	293,707	61,974	355,681	3.79	4.23	3.86
4	Negative Economic Impacts	193,399	30,670	224,069	2.50	2.10	2.43
5	Benefits of Climate Action	75,339	30,266	105,605	0.97	2.07	1.15
6	Positive Economic Impacts	1,157	1,949	3,106	0.01	0.13	0.03
Health Frame							
7	Health Frame Detection	106,777	21,164	127,941	1.38	1.45	1.39
8	Negative Health Impacts	106,777	21,164	127,941	1.38	1.45	1.39
9	Health Co-benefits	63,844	19,595	83,439	0.82	1.34	0.91
Security Frame							
10	Security Frame Detection	32,973	10,255	43,228	0.43	0.70	0.47
11	Resource Conflict	20,434	9,015	29,449	0.26	0.62	0.32
12	Climate-Driven Displacement	21,417	4,248	25,665	0.28	0.29	0.28
13	Military Disaster Response	0	10,253	10,253	0.00	0.70	0.11
14	Military Base Disruption	0	4	4	0.00	0.00	0.00
Justice Frame							
15	Justice Frame Detection	277,005	20,938	297,943	3.58	1.43	3.23
16	Unequal Access	215,729	4,791	220,520	2.78	0.33	2.39
17	North-South Responsibility	59,851	9,084	68,935	0.77	0.62	0.75
18	Winners & Losers	45,568	8,905	54,473	0.59	0.61	0.59
19	Intergenerational Justice	26,810	1,160	27,970	0.35	0.08	0.30
20	Unequal Impacts	13,216	2,188	15,404	0.17	0.15	0.17
Political Frame							
21	Political Frame Detection	2,411,443	409,831	2,821,274	31.13	28.00	30.63
22	Political Debate	2,360,465	401,652	2,762,117	30.47	27.44	29.99
23	Political Positioning	860,760	49,748	910,508	11.11	3.40	9.89

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Table B6 – *Continued from previous page*

#	Category	Count			Proportion (%)		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**
24	Policy Measures	205,010	18,833	223,843	2.65	1.29	2.43
25	Public Opinion	46,850	8,186	55,036	0.60	0.56	0.60
Scientific Frame							
26	Scientific Frame Detection	517,413	75,144	592,557	6.68	5.13	6.43
27	Discovery & Innovation	345,362	62,495	407,857	4.46	4.27	4.43
28	Scientific Controversy	212,955	14,929	227,884	2.75	1.02	2.47
29	Scientific Uncertainty	64,848	2,065	66,913	0.84	0.14	0.73
30	Scientific Certainty	48,588	0	48,588	0.63	0.00	0.53
Environmental Frame							
31	Environmental Frame Detection	206,579	74,554	281,133	2.67	5.09	3.05
32	Habitat Loss	107,529	62,541	170,070	1.39	4.27	1.85
33	Species Loss	118,604	49,797	168,401	1.53	3.40	1.83
Cultural Frame							
34	Cultural Frame Detection	290,549	82,465	373,014	3.75	5.63	4.05
35	Artistic Representation	202,492	44,568	247,060	2.61	3.05	2.68
36	Cultural Sector Footprint	13	82,382	82,395	0.00	5.63	0.89
37	Event Disruption	23,009	34,855	57,864	0.30	2.38	0.63
38	Loss of Indigenous Practices	7,201	18,778	25,979	0.09	1.28	0.28
PRIMARY CATEGORIES							
Actors/Messengers							
39	Actors/Messengers Detection	3,848,262	644,020	4,492,282	49.68	44.00	48.77
40	Public Official	1,286,959	223,649	1,510,608	16.61	15.28	16.40
41	Natural Scientist	279,864	70,656	350,520	3.61	4.83	3.81
42	Economic Expert	216,597	73,343	289,940	2.80	5.01	3.15
43	Cultural Expert	123,960	18,037	141,997	1.60	1.23	1.54
44	Activist	97,264	25,189	122,453	1.26	1.72	1.33
45	Health Expert	47,114	7,428	54,542	0.61	0.51	0.59
46	Social Scientist	32,278	19,609	51,887	0.42	1.34	0.56
47	Security Expert	19	8,916	8,935	0.00	0.61	0.10
48	Legal Expert	7	5,025	5,032	0.00	0.34	0.05
Events							

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Table B6 – *Continued from previous page*

#	Category	Count			Proportion (%)		
		EN	**FR**	**ALL**	**EN**	**FR**	**ALL**
49	Event Detection	1,421,264	277,560	1,698,824	18.35	18.96	18.44
50	Meeting/Conference	397,039	94,979	492,018	5.13	6.49	5.34
51	Policy Announcement	380,183	42,444	422,627	4.91	2.90	4.59
52	Publication	234,409	72,729	307,138	3.03	4.97	3.33
53	Extreme Weather Event	200,102	38,746	238,848	2.58	2.65	2.59
54	Election	64,663	28,706	93,369	0.83	1.96	1.01
55	Judiciary Decision	37,039	6,285	43,324	0.48	0.43	0.47
56	Cultural Event	964	22,998	23,962	0.01	1.57	0.26
57	Protest	7,741	5,293	13,034	0.10	0.36	0.14
Solutions							
58	Solutions Detection	1,822,433	307,710	2,130,143	23.52	21.02	23.13
59	Mitigation Strategy	1,261,603	201,352	1,462,955	16.29	13.76	15.88
60	Adaptation Strategy	48,963	18,119	67,082	0.63	1.24	0.73
EMOTIONAL TONE							
61	Neutral Emotion	4,879,832	1,057,425	5,937,257	62.99	72.25	64.46
62	Negative Emotion	2,975,222	513,870	3,489,092	38.41	35.11	37.88
63	Positive Emotion	634,048	184,697	818,745	8.18	12.62	8.89
GEOGRAPHIC FOCUS							
64	Canadian Context	3,387,282	573,263	3,960,545	43.72	39.17	43.00
URGENCY/ALARMISM							
65	Urgency/Alarmism	160,220	35,784	196,004	2.07	2.44	2.13
Total Sentences		**7,746,839**	**1,463,581**	**9,210,420**	**100.00**	**100.00**	**100.00**

Note: Counts represent the number of sentences in the CCF database annotated with each category. Proportions show the percentage of sentences containing each annotation. Categories are sorted by overall (ALL) prevalence within their hierarchical grouping. EN = English corpus (7.7M sentences), FR = French corpus (1.5M sentences), ALL = Combined corpus (9.2M sentences).

C Frame Dominance Analysis Details

C.1 Composite Dominance Score Components

Table 7 presents the detailed breakdown of dominance scores for each frame across the four analytical methods. Each method contributes 25% to the final composite score.

Table 7: Detailed frame dominance score components

Frame	**Info Theory** (25%)	**Network** (25%)	**Causality** (25%)	**Proportional** (25%)	**Composite** Score	**Rank**
Political	1.000	0.730	1.000	1.000	**0.932**	1
Economic	0.587	0.713	0.464	0.722	**0.622**	2
Scientific	0.705	0.348	0.841	0.465	0.590	3
Environmental	0.426	0.302	0.501	0.271	0.375	4
Social Justice	0.082	0.354	0.397	0.200	0.258	5
Cultural	0.099	0.348	0.206	0.232	0.221	6
Public Health	0.056	0.395	0.225	0.120	0.199	7
Security	0.000	0.500	0.000	0.000	0.125	8

Note: Dominant frames (shown in bold) determined by 3/4 consensus method

C.2 Information Theory Component Details

The information theory component (25% of the composite score) evaluates each frame's informational contribution through three complementary metrics that capture distinct aspects of information flow within the climate discourse system. The first metric, mutual information, quantifies the average information shared between each frame and all others in the corpus. This calculation employs discretized frame proportions using adaptive binning[12], where each frame's continuous proportion values are converted to discrete categories to enable information-theoretic analysis. For each frame, we compute its mutual information with every other frame and average these values, producing a measure of how much knowing about one frame's presence reveals about the broader discursive structure.

The second metric, entropy reduction, measures each frame's capacity to reduce uncertainty about the overall system when its state is known. This involves calculating the joint entropy of all frames together, then computing the conditional entropy of all other frames given knowledge of each specific frame[13]. Frames that substantially reduce system entropy when known are those that provide the most informational value for understanding the discourse, often because they tend to co-occur with specific configurations of other frames. The third metric, information content, assesses the self-entropy of each frame's distribution over time, measuring its temporal variability and unpredictability. Frames with high information

[12]The number of bins is determined adaptively as $n_{bins} = \min(10, \max(3, \lfloor\sqrt{N}\rfloor))$ where N is the sample size, ensuring appropriate granularity for datasets of varying sizes.

[13]Entropy reduction is calculated as $(H(X) - H(X|Y))/H(X)$ where $H(X)$ is the joint entropy of all frames and $H(X|Y)$ is the conditional entropy given frame Y.

content exhibit substantial variation in their presence across the corpus, while those with low information content maintain more consistent levels of presence.

C.3 Network Analysis Component Details

The network analysis component (25% of the composite score) captures frames' structural positions within the web of co-occurrence relationships, treating climate discourse as a complex network where frames are nodes and their co-occurrences form weighted edges. The analysis constructs temporal co-occurrence networks using 52-week sliding windows with 50% overlap, ensuring both temporal granularity and smooth evolution tracking. Within each window, edges are created between frames that co-occur above a 0.1 proportion threshold, with edge weights determined by the joint probability of co-occurrence[14]. This threshold ensures that only meaningful co-occurrences contribute to the network structure, filtering out noise from rare or incidental frame combinations.

Four metrics are extracted from these networks to assess different aspects of structural centrality. PageRank centrality identifies frames that serve as conceptual hubs, implementing Google's PageRank algorithm with a damping factor of 0.85[15]. This metric captures the principle that frames connected to important frames are themselves important, revealing which frames occupy central positions in the discourse network. Eigenvector centrality complements PageRank by measuring connectivity to other highly-weighted frames, identifying frames that associate with structurally central discourse elements. The temporal stability of both centrality measures is assessed through the inverse coefficient of variation across time windows, distinguishing frames with consistent network positions from those whose centrality fluctuates. Frames with high stability maintain their structural importance regardless of temporal variations in the discourse.

C.4 Causality Analysis Component Details

The causality analysis component (25% of the composite score) assesses frames' dynamic influence over the temporal evolution of other frames, identifying which frames act as drivers rather than merely correlates within the discourse. This analysis employs two complementary approaches to capture both linear and non-linear causal relationships. The first approach uses Granger causality testing through vector autoregression with adaptive lag selection[16]. For each frame pair, we test whether past values of one frame help predict future values of another beyond what the target frame's own history provides. The proportion of other frames that each frame Granger-causes indicates its breadth of causal influence, while the average F-statistic magnitude across significant relationships captures the strength of these causal connections.

[14]Edge weight between frames i and j is calculated as $w_{ij} = P(i > 0.1 \cap j > 0.1)$, the probability that both frames exceed the threshold in the same week.

[15]PageRank iteratively distributes importance scores where each frame's score is $PR(i) = (1-d)/N + d\sum_{j \in M(i)} PR(j)/L(j)$, with $d = 0.85$ as the damping factor, N the number of nodes, $M(i)$ the set of frames linking to i, and $L(j)$ the number of outbound links from j.

[16]Maximum lag is determined adaptively based on data length: 1 lag for $N < 10$ observations, 2 lags for $N < 20$, 3 lags for $N < 40$, and 4 lags otherwise. The Akaike Information Criterion selects the optimal lag within this range.

The second approach employs transfer entropy to detect non-linear information flow between frames that Granger causality might miss. Transfer entropy measures the reduction in uncertainty about a target frame's future state when we know both its past and the past of a potential source frame, compared to knowing only the target's past[17]. This information-theoretic measure captures directional information transfer without assuming linear relationships, revealing frames that systematically precede and potentially trigger the appearance of others. Frames with high transfer entropy scores export information to the broader discourse, suggesting they play initiating or agenda-setting roles in how climate change is discussed.

C.5 Proportional Dominance Component Details

The proportional dominance component (25% of the composite score) evaluates frames' sustained quantitative presence in the discourse through four metrics that capture different aspects of proportional representation. The first metric, mean proportion, simply measures each frame's average presence across the entire time period, providing a baseline measure of overall prevalence. However, since raw prevalence alone does not necessarily indicate dominance, three additional metrics assess the quality and consistency of this presence. The elevation score measures how much each frame's mean proportion exceeds the overall median of all frames, calculated as the relative difference normalized by the median[18]. This metric identifies frames that consistently rise above the general baseline of discourse.

The consistency metric, calculated as the inverse coefficient of variation, rewards frames that maintain stable presence rather than exhibiting high volatility[19]. Frames with high consistency scores are reliable components of the discourse rather than episodic appearances tied to specific events. The percentile rank metric positions each frame within the overall distribution of mean proportions, providing a normalized measure of relative prominence that is robust to outliers. Together, these four metrics distinguish frames that maintain substantial, elevated, and consistent presence from those that appear frequently but erratically or remain consistently present but at marginal levels.

D Additional Technical Details

D.1 Temporal Analysis Technical Specifications

The temporal evolution analysis employed several advanced techniques to track paradigmatic changes across multiple timescales. Annual dominance determination applied the complete four-component methodology to each year's data independently, with each year containing 52-53 weeks of frame proportions. The analysis pipeline first isolated each year's data, then calculated information theory metrics, network metrics, causality metrics, and proportional

[17] Transfer entropy from frame X to frame Y is calculated as $T_{X\to Y} = \sum p(y_{t+1}, y_t^k, x_t^l) \log \frac{p(y_{t+1}|y_t^k, x_t^l)}{p(y_{t+1}|y_t^k)}$ where y_t^k and x_t^l represent the past k and l values respectively.

[18] Elevation score is calculated as $\max(0, (p_i - m)/m)$ where p_i is frame i's mean proportion and m is the median proportion across all frames.

[19] Consistency is calculated as $1/(1+\sigma_i/\mu_i)$ where σ_i and μ_i are the standard deviation and mean of frame i's proportions over time.

dominance metrics using that year's data exclusively. Dominant frames were determined using the same consensus approach requiring agreement from at least three of four statistical methods: frames above median, natural break detection via maximum gradient, elbow method through cumulative variance curvature, and statistical outliers at mean plus 0.5 standard deviations.

Long-term trend extraction employed LOESS (Locally Weighted Scatterplot Smoothing) regression with adaptive bandwidth selection through cross-validation. The span parameter varied from 0.2 for recent years with dense data to 0.4 for early years with sparser coverage, optimizing the bias-variance tradeoff for each period. Confidence bands were constructed using residual bootstrap with 1,000 iterations to quantify uncertainty in trend estimates. Network evolution analysis constructed frame co-occurrence networks for overlapping three-year periods at two-year intervals, computing density as the proportion of possible edges present, modularity through Louvain community detection algorithm, average shortest path length between all frame pairs, and clustering coefficient measuring local network density. Paradigm concentration was tracked through two complementary indices: Shannon diversity index measuring evenness of frame distribution and the Herfindahl-Hirschman Index quantifying concentration analogous to market concentration metrics[20].

[20]The HHI formula $\sum_{i=1}^{n} p_i^2$ yields values from $1/n$ (perfect equality) to 1 (complete monopoly), with values above 0.25 indicating high concentration by antitrust standards.